



Master's Thesis

Deconvolution of Residence Time Distributions for Compartment Models via Convolutional Neural Networks

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Submitted at:

10.01.2025

Start Date. : 10.07.2024

Submission Date: 10.01.2025

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Acknowledgement

First and foremost, I would like to express my heartfelt gratitude to Prof. Dr.-Ing. Robert Güttel, thank you for giving me the opportunity to work on this challenging and stimulating research topic. I am deeply thankful for his insightful feedback and encouragement at key stages of my research.

I am incredibly grateful to my supervisors, M.Sc. Hannes Stagge and M.Sc. Max Gäßler, for their exceptional mentorship and unwavering support throughout this thesis. Their expertise, constructive feedback, and willingness to address my questions have been invaluable. I greatly appreciate their patience and trust in me, which enabled me to engage deeply with the research.

I owe enormous gratitude to my family for their constant support and encouragement throughout this journey. Their belief in me and unwavering motivation have been a continuous source of strength during this challenging yet rewarding experience.

I extend my deepest thanks to my best friend, who is truly like a sister to me. Her unconditional love and steadfast support have been a constant presence in my life. In moments of darkness, her infectious laughter has been a guiding light, and I am profoundly grateful for that.

I am also sincerely thankful to my close friends, Isil Altindas and Seval Cakır, whose support and encouragement have been invaluable. Their companionship and kind words have helped me persevere through difficult times, and their belief in me has motivated me to continue striving toward my goals. Thank you for always being there and for making this journey more meaningful with your presence.

Finally, I want to thank myself for the resilience and determination to keep growing, even in the face of storms. This journey has taught me the strength of perseverance, and for that, I am truly grateful.

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Nomenclature

Abbreviations

CTK	Chemical Transient Kinetics
CNN	Convolutional Neural Networks
D	Dimension
MS	Mass Spectroscopy
MSE	Mean Squared Error
nRTD	Neural Network Residence Time Distribution Model
PFR	Plug Flow Reactor
RTD	Residence Time Distribution
SNR	Signal to Noise Ratio
SSITKA	Steady-State Isotopic Transient Kinetic Analysis
TAP	Temporal Analysis of Product

Symbols

$\otimes$	-	Element-wise multiplication
Bo	1	Bodenstein number
c	Mol m^{-3}	Concentration
C	1	Normalised concentration
D	$\text{m}^2 \text{s}^{-1}$	Dispersion coefficient
E	1	Normalised residence time distribution function
F	1	Cumulative distribution function
G	1	Transfer function
J	1	Degree of mixing; number of cells in cascade
K	1	Number of samples in data set
k ₄	1	Kurtosis
L	m	Length
N	1	Number of tanks
N	1	Noise
Q	$\text{m}^3 \text{s}^{-1}$	Flow rate
q	$\text{m}^3 \text{s}^{-1}$	Lateral flowrate in each cell
R	1	Recycle
s ₃	1	Skewness
T	s	Time

u	s	Velocity
V	m^{-3}	Volume
x	1	Molar fraction
τ	s	Mean residence time

Greek Symbols

Δ	1	Delta
θ	1	Dimensionless time
δ	s^{-1}	Dirac delta function
β	1	Fraction in the active volume
α	1	Fraction of bypass flow
μ	1	Mean variance
$\sigma^2,$	s^{-1}	Peak variance
σ	1	Standard variance

Subscripts

i	In
o	Out
e	Kernel
p	Pulse
s	Step
m	Mixing
0	Initial
p	Plug
i	Number of convolutional layer
J	Model
Laminar	Laminar flow model
Dispersion	Dispersion model
Adler	Adler Hovorka model
Cholette	Cholette Cloutier
Unified	Unified time delay

Superscripts

$\hat{C}$	Prediction of quantity C
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$\hat{E}$	Prediction of quantity E
C'	Noisy data
$\hat{x}$	Prediction of quantity x

1 Introduction

The Residence Time Distribution (RTD) is a fundamental concept that provides critical insights into fluid dynamics [1]. It plays a vital role in understanding the behaviour of fluid elements, including their mixing efficiency, transport phenomena, and overall system performance. Accurately distinguishing the effects of fluid dynamics, as characterised by RTD, from kinetic processes is essential for identifying reaction kinetics and mechanisms in dynamic experimentation methods. However, despite its significance, RTD analysis remains a complex and time-intensive process.

Conventional RTD analysis involves an iterative process of qualitative and quantitative investigation, examining experimental data and predicting an appropriate model [2]. Compartmental models such as ideal plug flow reactors (PFR) or stirred tank reactors (STR) are used to represent systems in subdivisions [3]. While these models are widely used, they are limited by their reliance on idealised assumptions and lack the flexibility to deal with the complexities of real systems, such as dead zones and bypass flows as non-idealities. A further challenge is the determination of parameters, including the number of compartments and their specific characteristics, and fitting these models to experimental RTD data is rarely straightforward or accurate.

This study introduces the integration of Convolutional Neural Networks (CNN) into RTD analysis, aiming to replace conventional methodology by eliminating the iterative steps involved. CNN is a powerful feedforward neural network that learns patterns directly from the data through kernel optimisation, enabling it to generate accurate predictions without intermediate steps [4]. This inherent capability, coupled with the mathematical equivalency between CNN and convolutional analysis for RTDs, constitutes the foundation of the developed Neural Network Residence Time Distribution (nRTD) model. Hence, the nRTD model offers a rigorous, data-driven solution which can be applied to various non-ideal flow scenarios without oversimplification.

In order to validate the predictive accuracy of the nRTD model, it is trained on synthetic data sets generated from a wide range of theoretical compartment models, from simple laminar flow to advanced models such as Cholette and Cloutier, Adler-Hovorka and the Unified Time Delay model. This rigorous approach ensures that the nRTD model can effectively capture both ideal and non-ideal system behaviour, providing a powerful tool for real-world applications where conventional methods often fail. Unlike traditional RTD methods that rely on predefined models and extensive manual tuning, the nRTD framework learns directly from the data and adapts to different flow scenarios and non-idealities. To address more complex systems with multiple sub-compartment RTDs, a multilayer nRTD model is developed as an extended methodology to facilitate compartmentalisation. As a proof of concept for the multilayer design, several

literature compartment models are implemented in a cascading workflow, demonstrating the ability of the model to represent individual compartments within a complex system. The capability of the multilayer nRTD is investigated using experimental data from a realistic test rig setup designed for dynamic experiments. The prediction accuracy remains successful for the representation of individual compartments presented despite imperfections in the experimental setup related to perturbations and failures.

2 Theoretical Background

The analysis of RTD provides key knowledge to determine the transport of fluid elements through a reactor with an understanding of the mixing that affects product quality, conversion rates, and overall reactor performance [5]. It plays a critical role in various dynamic experimentation methods where systems are operated under changing conditions that induce variations in behaviour affecting reaction kinetics and product formation. Understanding the residence time distribution is essential for studying the dynamics of kinetic processes in a range of methodologies, including periodic transient kinetics (PTK), steady-state isotopic transient kinetic analysis (SSITKA), chemical transient kinetics (CTK), and temporal analysis of product (TAP) experiments.[6,7] Consequently, the decoupling of flow effects from the intrinsic reaction kinetics is critical for the reliable determination of the true kinetic parameters and reaction mechanism of the underlying chemical processes.

2.1 Conventional RTD analysis

Three main stages can be considered for the conventional methodology of RTD analysis: (i) nonchemical experimentation, (ii) qualitative analysis, and (iii) mathematical modelling [2]. Pulse injection and step tracer experiments are used in the experimental phase, measuring the system response to determine RTD. In a pulse experiment, a tracer with a small concentration, $C_{in}(t)$, is suddenly injected as a single shot into the feed stream entering the reactor in an infinitely short amount of time, ideally approximating a Dirac delta function, $\delta(t)$ [8]. Although the pulse input offers detailed insight, it can be challenging to achieve in practice. The profile of the input function has a spike at the inlet, followed by a rapid decrease, as illustrated in Figure 2.1.a. Alternatively, step experiments involve altering the operating conditions of the reactor, typically the flow rate or concentration of a reactant, and measuring the response of the system over time using a tracer. For a step input, the tracer concentration is suddenly changed from zero to a constant value of $C_0(t)$, represented by Equation (2.1), where $H(t)$ is the Heaviside step function.

$$C_{in}(t) = C_0 H(t) \quad (2.1)$$

Figure 2.1.b illustrates the instantaneous increase in the step function at time zero. While pulse and step experiments are relatively simpler to interpret, periodic and random experiments are also potential approaches. In the case of periodic input, the input exhibits a sequence which is repeated and orderly, while random input does not demonstrate any pattern, as shown in Figures 2.1.c and 2.1.d, respectively. The decision of which input method to utilise is based on the dynamics of the experiment and the information that is being targeted about the system under investigation. Also, the selection of an appropriate tracer is a critical determinant of

accurate RTD measurements [1]. Ideally, the tracer should have similar flow properties to the target substance. The tracer must be detectable at very low concentrations and the instrumentation must be preferably sensitive. Key considerations include the physical properties of the tracer, such as density and viscosity, and chemical inertness to avoid system alteration, as well as stability under operating conditions.

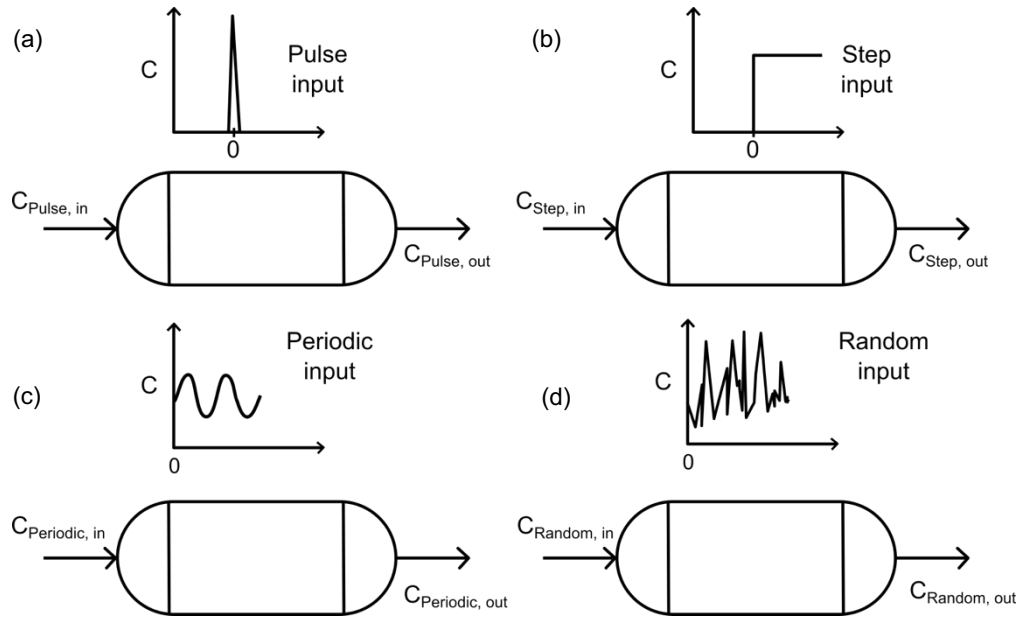


Figure 2. 1: Different input methodologies for investigating the flow pattern.

As a second step, qualitative analysis is carried out on the resulting information from the concentration and time data, and functions are utilised at this stage to evaluate the experimental data to determine the flow and mixing behaviour. Concentration-time profiles are utilized to analyse the overall behaviour of the system. These profiles offer valuable insights into how the reactor responds to the input functions. Figure 2.2.a and 2.2.b illustrate typical system responses to step and pulse inputs, respectively.

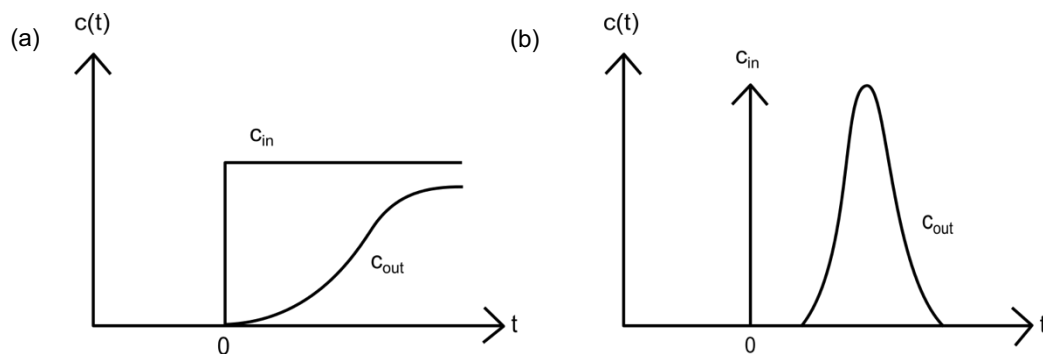


Figure 2. 2: System response to (a) step and (b) pulse input.

In Figure 2.2.a, a steep sigmoid curve characterises the response to a step input. The concentration initially remains at the baseline level, followed by a rapid increase, and finally

approaches the steady-state value of the input. Furthermore, Figure 2.2.b illustrates the response to a pulse input, which typically results in an asymmetric peak. The shape and rate of this transition provide information about the mixing characteristics of the system and possible non-idealities [5]. While the graphical representations provide valuable qualitative insights, the analysis of the data requires the introduction of specific functions and parameters to enable a thorough mathematical handling.

The RTD function, denoted as $E(t)$, describes the temporal dynamics of the different fluid elements in the reactor [1]. The $E(t)$ function, which is also known as exit age distribution, describes the fraction of fluid elements exiting the reactor with a residence time between t and $t + dt$. The function can be derived directly from the outlet tracer concentration for pulse experiments, with units of t^{-1} . Differently, for step inputs, it is defined by the cumulative distribution function, $F(t)$, obtained by integrating $E(t)$ over a series of time intervals [9]. The $E(t)$ function is typically derived from pulse inputs, while $F(t)$ is obtained from step inputs due to the time interval characteristics of the experiments [10]. The description of $E(t)$ for pulse and step input is given by Equations (2.2) and (2.3), respectively.

$$E_p(t) = \frac{C(t)}{\int_0^\infty C(t) dt} \quad (2.2)$$

$$E_s(t) = \frac{dF(t)}{dt} \quad (2.3)$$

Furthermore, $F(t)$ represents the cumulative contributions of tracer elements that have resided in the system for a time less than or equal to time before exiting. The relationship between $F(t)$ and $E(t)$ is described by Equation (2.4), where $F(t)$ is obtained by integrating $E(t)$ over time [11,12].

$$F(t) = \int_0^t E(t) dt \quad (2.4)$$

In order to gain further analysis of behaviour and flow patterns of the system, the central moments of the RTD can be investigated to achieve additional information regarding the properties of the functions. The mean residence time, represented by the first central moment of the RTD, is the average time that fluid elements spend in a system and the central tendency of the RTD is calculated using Equation (2.5) [5].

$$\tau = \int_0^\infty t E(t) dt \quad (2.5)$$

The second central moment, variance, σ^2 , measures the spread of residence times, with higher values indicating more intense mixing, and thirdly skewness, s_3 , assesses the asymmetry of

the distribution. Finally, kurtosis, k_4 , quantifies the peakiness of the distribution, where higher values indicate sharper peaks and narrower distributions [2].

The qualitative phase is followed by the implementation of mathematical models designed to simulate and predict fluid flow and mixing dynamics. Both of these stages involve a cyclical process of iteratively identifying a model, implementing it, and assessing its accuracy through evaluation. It is likely that the system is initially associated with an inaccurate model and a scenario that often arises after the application and comparison of the behaviour of the system.

2.2 The Convolution Integral

In practice, ideal pulse or step inputs are rarely achievable, resulting in deviations that can significantly affect the accuracy of RTD measurements [9]. These variations in the input signals can propagate through the system, leading to misinterpretation of the reactor behaviour. Therefore, the convolution integral is often used to enable better prediction of the system response to non-ideal step or pulse inputs [13]. Moreover, convolution is a binary operation that combines two functions, representing the input stimulus and the system characteristic. The resulting function is generated by integrating the pointwise multiplication of the original functions. In a manner of words, convolution "blends" one function with another to model the response of a system to an input signal [13]. The mathematics of the concept of convolution is rigorously defined as the integration over the entire spatial domain of a function, $f(x)$, multiplied by another function $g(x)$, evaluated at the displacement, $u - x$. This integral spans the variable, x , and overall dimensions, usually from negative infinity to positive infinity, which is particularly applicable in multidimensional cases [11]. Thus, it leads to a new function that depends on a new variable, u , exemplified by the formalized equations that explain the convolution process, as given in Equation (2.6). Due to the property of commutativity in convolution integrals, the displacement is applicable. Therefore, the order of the functions to be interchanged without changing the result, as shown in Equation (2.7). Thus, flexible analysis of the system is possible.

$$C(u) = f(x) \otimes g(x) = \int_{\text{space}} f(x) g(u-x) dx \quad (2.6)$$

$$= g(x) \otimes f(x) = \int_{\text{space}} g(x) f(u-x) dx \quad (2.7)$$

However, in chemical reactor design, where time is the independent variable, the lower limit of the integration is often set to zero and the upper limit of the integration is defined by a specific time, t , when computing the convolution [13]. Additionally, τ , as given in Equation (2.8), can be

used in the description of the convolution process to obtain a better indication of how long the reactants spend in the reactor, which is critical for predicting the output of the system.

$$C(t) = \int_0^t f(\tau) g(t-\tau) d\tau \quad (2.8)$$

When integral convolution is applied to RTD analysis in compartments, given a known input signal and RTD, the output signal is calculated as shown in Equation (2.9). This process involves determining the times associated with both the input and output signals. Accordingly, Figure 2.3 illustrates the possible representation of the convolution application.

$$C_{out} = E \otimes C_{in} \quad (2.9)$$

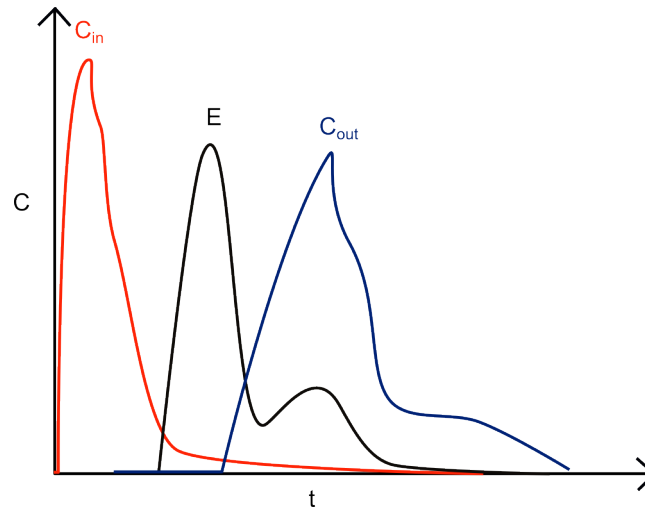


Figure 2. 3: The representation of an convolved input, C_{in} , with a signal, E to provide output, C_{out}

Many practical applications involve multiple compartments connected in series, which creates a cascade of modulations in the tracer signal. As the signal passes through each compartment, it is transformed according to the specific RTD of that compartment [14]. This sequential modification of the signal adds layers of complexity to the analysis, making it challenging to predict the system's overall behaviour accurately. Furthermore, it allows for investigating a specific unit within a multi-unit system, as described in Equation (2.13). [13]. In order to determine the unknown E curve of a compartment, inversion of the convolution, called deconvolution, should be implemented.

$$C_{out,1} = E_1 \otimes C_{in} \quad (2.10)$$

$$C_{out,2} = E_2 \otimes C_{out,1} = E_1 \otimes E_2 \otimes C_{in} \quad (2.11)$$

$$C_{out,3} = E_3 \otimes C_{out,2} = E_1 \otimes E_2 \otimes C_{in} \quad (2.12)$$

$$C_{out,i+n} = E_{i+n} \otimes C_{out,i+n-1} = E_1 \dots \otimes E_{i+n} \otimes C_{in} \quad (2.13)$$

Deconvolution enables the determination of the RTD function from the injected and detected tracer concentration profiles, even in cases where non-ideal tracer inputs are used. This process is beneficial in case a direct measurement of the RTD is not feasible or interested on an analysing individual compartment in a system. Although analytical deconvolution is not achievable, numeric approximations are available, namely the flow model fitting [15].

A variety of mathematical RTD models are used to describe the flow and mixing behaviour in systems, providing an accurate representation of the system being analysed [1]. The choice of an appropriate model depends on several factors, including vessel geometry and flow rate. The modelling process becomes more complex when chemical reactions are included, as it can also incorporate reaction kinetics to predict conversion and yield. The overall complexity emphasizes the importance of a goal-oriented approach: collecting all relevant data, distinguishing between critical and non-critical details, and finding a balance between simplicity and comprehensiveness [16].

2.3 Nuclei elements

Nuclei elements serve as fundamental components in the study of fluid dynamics, particularly in understanding the behaviour of fluids in various systems. In order to simplify calculations and predict reactor performance, two idealised models, the plug and laminar flow, represent extreme cases of flow behaviour with uniform mixing of the system contents on entry [2]. Figure 2.3 illustrates these nuclei, which include the ideal flow and mixing models alongside the non-ideal flow models [3].

2.3.1 Ideal flow models

The plug flow model assumes that the fluid velocity is constant along each pipe cross-section, even though fluid flow and mixing perpendicular to the pipe axis cannot be considered separately in the real world [17]. The uniformity of the velocity in a dedicated cross-sectional area is shown in Figure 2.4. a. In addition, the plug flow model assumes that a system is completely filled with fluid, where the diffusion terms are negligible, and radial gradients are ignored with the absence of mixing of earlier fluid with later arriving fluid, back mixing, eddies or vortices. In laminar flow, fluid velocity progressively increases toward the centre, resulting in a parabolic velocity profile, as shown in Figure 2.4.b. As a result, the fluid elements in a

laminar flow model experience different residence time: those in the centre exit the system first, while those near the wall leave last [1].

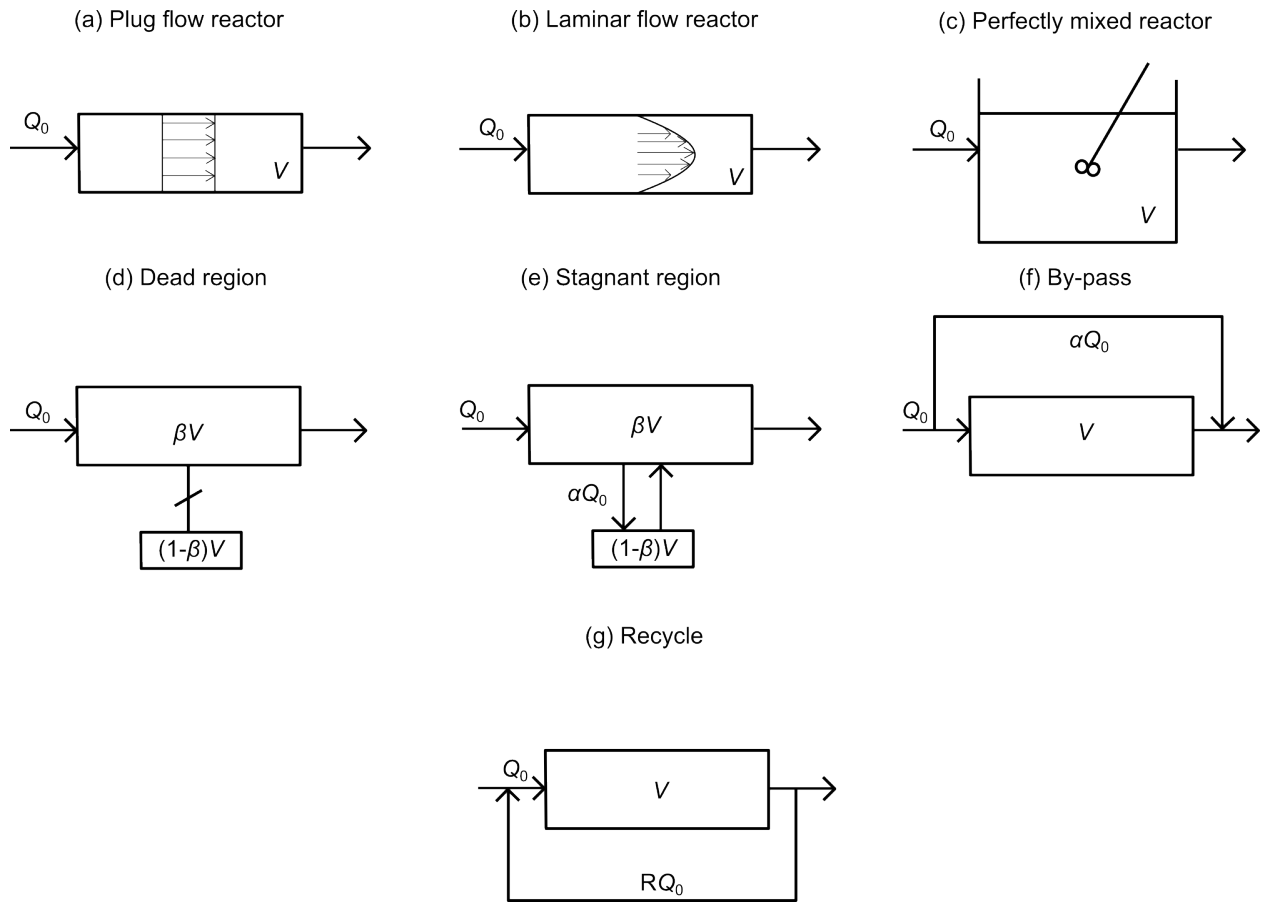


Figure 2. 4: Schematic representation of the nuclei elements of compartment models [3]

2.3.2 Complete mixing model

The model assumes that the fluid elements within a system are always perfectly mixed, therefore, it is shown as an ideal STR in Figure 2.4.c [5]. It implies that all properties, such as temperature, concentration, and density, are uniform throughout the reactor at all times. The model assumes that incoming fluid immediately and uniformly mixes with the existing contents of the system as it enters, resulting in a homogeneous mixture. Consequently, the properties of the fluid inside the reactor are identical to those in the outlet flow.

2.3.3 Deviation from the ideal flow

In real systems, fluids commonly exhibit complex flow patterns and non-uniform mixing that can be difficult to predict. Several factors cause this behaviour to deviate from ideal models, including the geometry of the vessel, the type, size and placement of the inlet and outlet pipes, the type of mixer used, the properties of the fluids, and the flow rate. While the primary

performance of a system is mainly determined by its active zones, the overall system response is significantly affected by deviations from ideal conditions under non-ideal flow conditions [2,3]. These deviations include dead zones, short circuits and recirculation. Dead zones are areas where the portion of the system is not filled, indicating unutilised system capacity, as shown in Figure 2.4.d. Similarly, regions where the fluid is stationary, as shown in Figure 2.4.e, are referred to as stagnant regions. In addition, bypass characteristics occur when fluid avoids certain areas of the system, resulting in non-uniform residence times; this phenomenon is shown in Figure 2.4.f. In some cases, due to internal recirculation, portions of the fluid may circulate through the same compartment multiple times, the altering of flow dynamics by introducing additional flow paths is displayed in Figure 2.4.g [1].

Figure 2. 5 illustrates the effect of the non-ideal flow elements on the ideal plug flow model and how it deviates regarding the RTD function [1]. Figure 2.5.a represents the ideal pulse input with volume, V , and flow rate, v , as an infinitely high peak with a narrow E -curve profile. Since the presence of a dead volume, V_d , reduces the actual volume of a reactor, the profile of the $E(t)$ function results in an earlier breakthrough of a step sample introduced at the reactor inlet, as shown in Figure 2.5.b Hence, the dead volume effectively shortens the effective path of the fluid, leading to premature tracer emergence depending on the volume fraction still actively flowing through the reactor [18]. Meanwhile, in by-pass, as shown in Figure 2.5.c, part of the tracer will break through immediately after tracer injection, v_b , and the area under the pulse will decrease due to the change in the flow rate of the pulse. The remaining flow leaves the reactor with a corresponding delay.

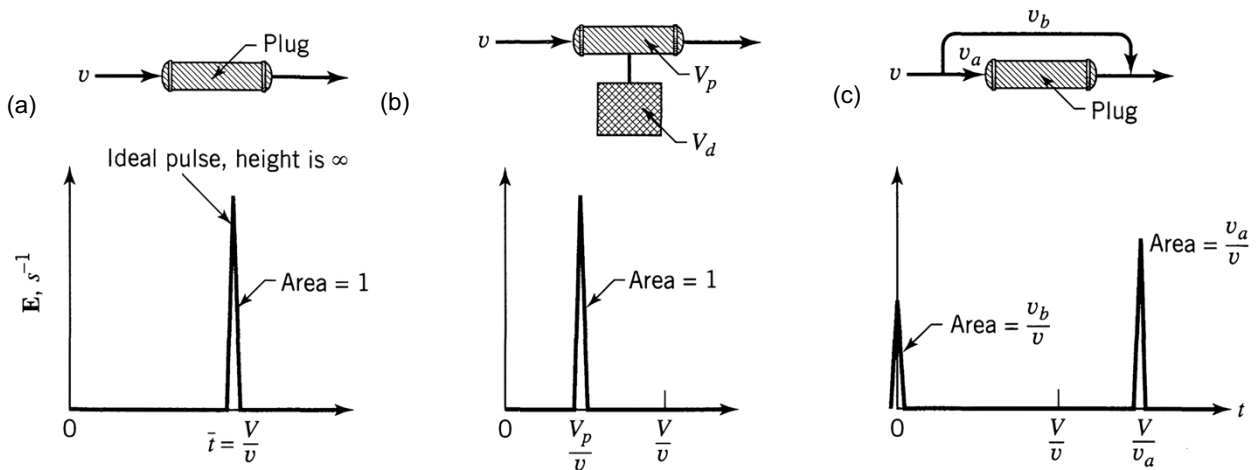


Figure 2. 5: Schematic representation of the RTD function for (a) ideal plug flow model, (b) ideal plug flow model with dead zone, and (c) ideal plug flow model with bypass, each with pulse input. [1]

The complete mixed model incorporating non-ideal flows, in particular dead volume and bypass, exhibits characteristics in its RTD function as shown in Figure 2.6. In the presence of dead volume, illustrated in Figure 2.6.b, the effective reactor volume decreases, resulting in a modified RTD function characterised by a narrower distribution with a shorter tail. Conversely,

in the presence of bypass flow (Figure 2.6.c), the profile has a steep initial slope, representing the fraction of fluid bypassing the main reactor volume, while the tail resembles that of an ideal STR, corresponding to fluid undergoing normal mixing. The bypass flow produces a broad range distribution in the RTD, with an early peak indicating bypassed fluid and a later, more dispersed peak corresponding to normally mixed fluid[2].

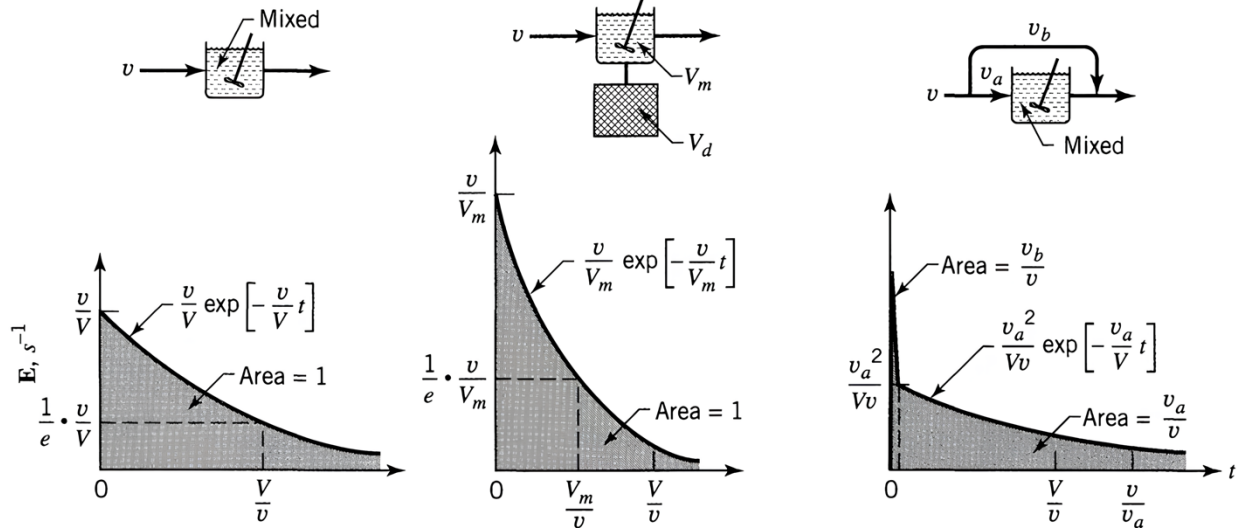


Figure 2. 6: Schematic representation of the RTD function for (a) ideal STR, (b) ideal STR with dead zone, and (c) ideal STR with bypass.[1]

2.4 Compartment models

Compartment models divide the system into a series of interconnected compartments through which the fluid flows. Rodrigues proposed that compartment models can be developed by assembling the nuclei elements (Section 2.3) into more advanced models [3]. The models can be divided into two categories, basic and complex, according to their number of nuclei and complexity. Therefore, the models that have more than one nucleus are referred to as complex compartment models.

2.4.1 Basic Flow Models

2.4.1.1 Dispersion model

In order to investigate and define the small deviations from ideal plug flow in real reactors, a dispersion model can be utilized. This model is applicable to turbulent flow in pipes, laminar flow in very long tubes, as well as flow in packed beds and long channels. The dispersion model accounts for the broadening of the flow distribution due to axial mixing, in contrast to ideal plug flow, which assumes a uniform radial distribution of fluid velocity. It treats the flow as an ideal plug flow that is superimposed with a diffusion-like process, leading to axial mixing without stagnant pockets or significant bypassing or short-circuiting of the fluid within the vessel [1]. Considering an ideal pulse input is introduced into the vessel, the spreading of the pulse

as it goes through the system can be characterised by the dispersion coefficient, D , with the assumption of axial back-mixing through a diffusion-like process.

The Bodenstein number, Bo , gives a dimensionless indication of the relative significance of convective transport, u , compared to diffusive transport, D_{ax} , in the fluid flow, while the dispersion coefficient quantifies the degree of axial back mixing [19]. The relationship is described by Equation (2.14), which illustrates the extent to which Bo encompasses the balance between these two transport mechanisms.

$$Bo = \frac{uL_R}{D_{ax}} \quad (2.14)$$

The Bo number is an important parameter for characterizing flow behaviour in reactors, ranging from ideally mixed to plug flow conditions. When Bo equals 0, it indicates complete back mixing, which is the ideal state for a continuous STR. At low Bo values, the reactor demonstrates significant axial dispersion, behaving similarly to a fully mixed system with broad residence time distributions. As Bo increases, the flow transitions toward plug flow characteristics, resulting in narrower RTD profiles and reduced back mixing. High Bo values (≥ 100) suggest minimal axial dispersion, closely resembling plug flow behaviour with narrow, symmetrical RTD. In the extreme case where Bo is infinite, there is no back mixing, and the system behaves like an ideal flow channel with continuous flow [20].

The corresponding E -function of the compartment model, can be derived from the Danckwerts' solution to the second-order differential equation [21]. This approach describes an unsteady-state material balance in an open system, where concentration continuity is maintained at the boundaries. The Equation (2.15) represents the dimensionless form of the RTD density function for the dispersion model, which depends on the Bo and dimensionless time, θ [19].

$$E_{\theta}(\theta) = \frac{1}{2} \sqrt{\frac{Bo}{\pi\theta}} \exp \left[-\frac{Bo(1-\theta)^2}{4\theta} \right] \quad (2.15)$$

2.4.1.2 Tank in series model

Comparable to the dispersion model, the tank-in-series model can be applied to define a compartment by using similar assumptions and methodologies. The model is straightforward and it can be used with any kinetics and relatively easily extended to any arrangements of compartments, with or without recycling stream. For the material balance of N number of tanks, according to the derivation of MacMullin and Weber, gives Equation (2.16). [1]

$$E_{\theta}(\theta) = N \frac{(N\theta)^{N-1}}{(N-1)!} e^{-N\theta} \quad (2.16)$$

The number of tanks in series, which reflects the degree of mixing within the system, can be calculated using Equation (2.17) with the dimensionless variance, σ_θ^2 . Furthermore, when N equals 1, the system yields an STR characterised by perfect mixing and a wide RTD profile. However, the flow behaviour shifts towards plug flow as N increases and even for larger values of N , the system approaches ideal plug flow conditions, resulting in a narrower and more defined RTD.

$$N = \frac{1}{\sigma_\theta^2} \quad (2.17)$$

2.4.1.3 Convection model for laminar flow

In viscous fluids, radial diffusion between fluid particles flowing at different velocities can often be negligible due to the inherent viscosity of the fluid. Thus, laminar flow predominates, characterized by a distinctive parabolic velocity profile. This flow regime is governed by high momentum diffusion and low momentum convection, resulting in fluid particles moving in smooth and parallel layers. Ideally, fluid elements are assumed to move past each other without molecular diffusion interactions. Consequently, the distribution of residence times is attributed solely to velocity variations in the flow profile. The method of introducing the tracer into the flowing fluid and the technique of measurement strongly influence the shape of the response curve [1,2]. The implementation of the convection for laminar flow model should take into account the particularities of the experimental setup. Therefore, the concepts of flux and planar inputs/outputs are crucial for implementing the model and understanding the flow behaviour. In flux introduction, the tracer is introduced proportionally to velocity, resulting in a higher concentration at the centreline of the pipe and minimal tracer near the walls. In contrast, planar introduction ensures a uniform distribution of tracer throughout the pipe, achieved by multiple injectors or a flash of light on a photosensitive liquid. In terms of measurement, flux measurement involves a mixing cup approach, where all the fluid exiting the tube is collected to determine the product of velocity and concentration. On the other hand, planar measurement involves through-the-wall methods, such series of probes distributed across the pipe. And provides an instantaneous measurement of the average concentration at a given time. The combination of flux and planar inputs/outputs in an experimental setup can vary, resulting in different flow characteristics and residence time distributions. Therefore, different combinations can be used to combine flux and planar inputs and outputs in an experimental setup. Equation (2.18, 2.19, and 2.20) show the E function for flux-flux, flux-planar and planar-planar combinations, respectively.

$$E = \frac{\tau^2}{2t^3} \text{ for } t \geq \frac{\tau}{2} \quad (2.18)$$

$$E = \frac{\tau}{2t^2} \text{ for } t \geq \frac{\tau}{2} \quad (2.19)$$

$$E = \frac{1}{2t} \text{ for } t \geq \frac{\tau}{2} \quad (2.20)$$

2.4.2 Complex Flow Models

A more complex model to describe the residence time behaviour of a real STR has been proposed by Cholette and Cloutier [3,22]. The model considers a by-pass stream, that is not mixed with the reactor contents and a dead volume that does not exchange with the ideally mixed remaining volume of the reactor, as shown in Figure 2.7.

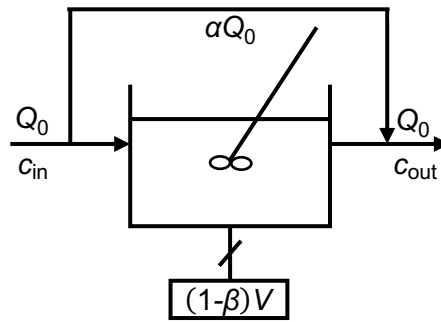


Figure 2. 7: The Cholette and Cloutier compartment model representation.

The transfer function $G(s)$ is a mathematical representation that describes the relationship between the input and output of a dynamic system in the frequency domain. The bypass flow, αQ_0 , and a dead zone with a volume of $(1-\beta)V$ relative to residence time, $\tau = V/Q_0$ are the main three parameters to describe $G(s)$, of the Cholette and Cloutier model, Equation (2.21).

$$G(s) = \alpha + (1-\alpha) \frac{1}{1 + \frac{\beta\tau}{1-\alpha}s} \quad (2.21)$$

The RTD function can be derived from the $G(s)$ by the inverse Laplace transform, which can be expressed mathematically as shown in Equation (2.22). Applying the inverse Laplace transform to $G(s)$ yields $E(t)$ which reflects the time dependent behaviour of the system with the $\delta(t)$, Equation (2.23) [3].

$$E(t) = L^{-1}G(s) \quad (2.22)$$

$$E(t) = \alpha\delta(t) + (1-\alpha)\frac{(1-\alpha)}{\beta\tau}e^{-\frac{(1-\alpha)t}{\beta\tau}} \quad (2.23)$$

Compartment models typically comprise multiple sub-compartments that collectively define the system. The Adler and Hovorka model feature a PFR with a volume of V_P and a perfectly mixed STR with a volume of V_A [3,23]. These components exchange mass with a stagnant zone of volume V_m , all of which are integrated into an actual reactor with a total volume of V and a flow rate of Q_0 , as illustrated in Figure 2.8.

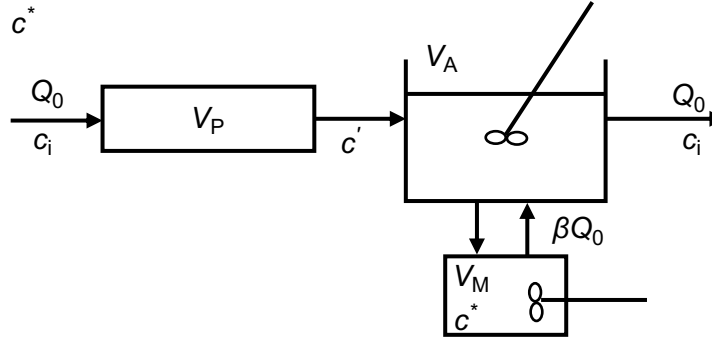


Figure 2. 8: The Adler and Hovorka compartment model representation.

In order to define the flow of material between compartments, the term of βQ_0 is introduced and the transfer function is derived as shown in Equation (2.24). Since the characteristics of the residence time are unique to each compartment, specific terms are included in the equation of the transfer function to reflect these differences. The residence time of the PFR, STR and stagnant zone is represented as τ_p , τ_a , and τ_m , respectively. If $\beta = 0$, absence of stagnation zone, the system yields into a Kramer system consisting of a PFR and a perfectly mixed reactor in series [3]. The derivation of $E(t)$ follows in Section (3.3), but is introduced in this section for completeness.

$$G(s) = \frac{e^{-\tau_p s}}{\left(1 + \beta + \tau_a s - \frac{\beta}{1 + \tau_m s}\right)} \quad (2.24)$$

Levich et al. consider hydrodynamic mixing in a porous medium through a model that includes a series of perfect mixers with stagnant zones [24,25]. This model effectively captures the complexities of fluid behaviour in such systems, highlighting the significant impact of stagnant zones on mixing efficiency and overall reactor performance. Subsequently, the unified time delay model, which consists of a packed bed structured in discrete cells, is illustrated in Figure 2.9. The total volume of the central flow region is denoted as V while J represents the number of cells in the series. The model conceptualizes the reactor as a series of discrete, well-mixed

cells within the main flow region, Q_0 . Each cell accounts for both Q_0 and lateral flow, q , dynamics. Additionally, the q from each cell, which is proportional to its length is described by the transfer function $g_D(s)$. Consequently, Equation (2.25) illustrates the time-delay distribution in the lateral zones. The residence time in the main flow is represented by τ_s , while τ_D accounts for the delay in residence time due to lateral flow. Consequently, to capture the dynamics of lateral flow and the associated delays in the overall system behaviour, the transfer function is defined in Equation (2.26) [25]. The derivation of $E(t)$ is presented in Section (3.3).

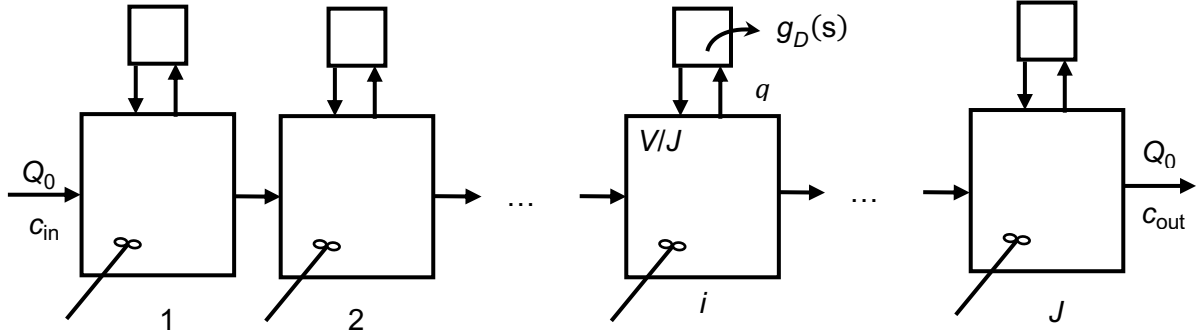


Figure 2. 9: Unified time delay model for dispersion in flowing media compartment model representation.

$$g_D(s) = \frac{1}{1 + \tau_D s} \quad (2.25)$$

$$G(s) = \left\{ 1 + \frac{1}{J} (\tau_s + \alpha - \alpha g_D(s)) \right\}^{-J} \quad (2.26)$$

2.5 Convolutional neural network

Even though the convolutional integral is considered beneficial for predicting system responses in analysis, inherent variations within systems introduce a level of error into the results and the struggle to choose an appropriate model is still a concern [16]. Therefore, Convolutional Neural Network (CNN) techniques are considered to be a promising solution to overcome the shortcomings of the traditional RTD methodology. CNNs are capable of performing successfully in computer vision tasks of all kinds, including object identification, image recognition, and image retrieval [4]. They utilize deep learning by applying multiple convolutional layers that autonomously learn hierarchical representations from input data. A typical CNN consists of several interconnected steps, as shown in Figure 10. It starts with an input layer, followed by one or more convolutional layers that extract features from the data. These are followed by pooling layers to downsample the feature maps and reduce dimensionality. The extracted features are then passed through a fully connected layer, which learns complex patterns. Finally, the network ends with an output layer that generates predictions or classifications based on the learned features.

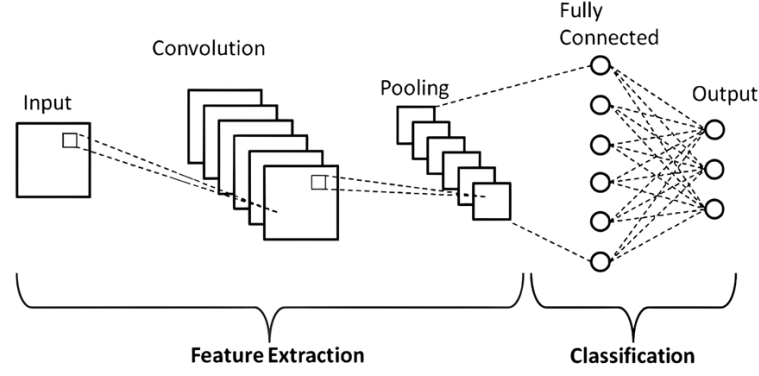


Figure 2. 10: Schematic representation of the architectural layers of CNN [26].

The input to the convolutional layer is defined by its size (N, C_{in}, L) , where N represents the batch size, indicating the number of input samples processed simultaneously. Each sample contains C_{in} , the number of input channels. The overall process of a CNN can be divided into two segments: the forward pass and the backward pass. In the forward pass, input data, is sequentially processed through the network's layers to generate predictions, with each layer in the CNN, characterized by its weight. Weight represents the learnable kernel applied between the k^{th} C_{in} and the j^{th} output channel, C_{out} . The layers are responsible from extracting essential spatial and temporal patterns that are crucial for understanding system behaviour over time. They process the input and produce an output that serves as the input for the subsequent layer. The output of the layer has dimensions (N, C_{out}, L) where is the number of C_{out} , corresponding to the feature maps produced by the convolutional kernels. A convolution layer consists of a convolution kernel that computes feature maps. The bias is an optional scalar term added to the output of the j^{th} channel to adjust its value for enhancing the model's ability to learn. In the case of multiple convolution layers, different kernels can be implemented, each capturing distinct aspects of the input.[27] Specifically, each neuron in a feature map is connected to a localized region of neurons from the previous layer, known as its receptive field. The new feature map is obtained by first convolving the input with a learned kernel and then applying an element-wise nonlinear activation function to the convolved results. After the forward pass generates a prediction, it is compared to the target parameter corresponding to the input in the loss layer, resulting in a loss. The backward pass, or backpropagation, then commences to optimize the network's parameters. During this phase, gradients are computed for each layer, representing the partial derivatives of the loss with respect to both the layer parameters and its input. This gradient computation propagates backwards through the network, allowing for the adjustment of parameters to minimize the loss function. The overall sequence can be explained by Equation 2.27 where L specifies the length of the signal sequence, representing the spatial or temporal resolution of the input data [28].

$$\text{out}(N_i, C_{\text{out},j}) = \text{bias}(C_{\text{out},j}) + \sum_{k=0}^{C_{\text{in}}-1} \text{weight}(C_{\text{out},j}, k) * \text{input}(N_i, k) \quad (2.27)$$

The Equation (2.27) represents the valid cross-correlation operator describes the process of sliding the kernel over the input data and computing element-wise multiplications followed by summation. Cross-correlation and convolution are mathematically equivalent under the appropriate kernel transformation [29]. The difference lies in the orientation of the kernel, where convolution involves flipping the kernel before the signal traverses through, whereas cross-correlation does not. Consequently, with proper kernel flipping, cross-correlation operations can be accurately analysed within the convolution framework.

Optimization algorithms are essential for efficient learning and convergence in deep learning. The Adaptive Moment Estimation (ADAM) optimizer is among the most favoured selections [30]. It is an iterative optimization algorithm used to minimize the loss function during the training of neural networks. It takes advantage of momentum by using the moving average of the gradient instead of the gradient itself while also adapting the learning rate for each parameter based on the first and second moments of the gradients. Moreover, several hyperparameters are used in machine learning techniques to influence the accuracy of the profile fitting in neural networks. They include the learning rate, which controls the magnitude of parameter updates during optimization, and the epoch number, which represents the cumulative number of iterations over the entire data set during training [31-33].

CNN and RTD fitting share a fundamental similarity: both involve the identification of an unknown parameter that transforms input data to produce a desired output. In CNN, the kernel represents a learnable filter that extracts meaningful features from the input data. It is initially undefined and is iteratively optimised during training to minimise the discrepancy between predicted and actual outputs. Similarly, in RTD fitting, the objective is to identify the RTD function analogous to the kernel in CNN, which produces an output that matches the observed data when convolved with the input. This common framework for kernel identification highlights the inherent alignment between CNN and RTD fitting, making CNN an ideal computational tool to address the challenges of RTD analysis. By employing its ability to learn from data, CNN provides a systematic and adaptive approach to modelling RTD, eliminating the need for predefined functions and extensive manual parameter tuning.

3 Experimental Setup and Method

3.1 Experimental setup

The experimental framework utilised in this study is based on the experimental setup used in the PTK experiments as described by Meyer et al [34]. The detailed experimental procedures, chemicals used, and other details are provided in the previous work [35], but it is important to outline the general setup with the corresponding experimental approach. The experimental design is configured to characterise RTD profiles in different compartments of the system. Therefore, the experimental setup is segmented into several subsections that are essential for profile characterisation within each compartment, as shown in Figure 3.1. The segmented approach allows the flow dynamics and mixing characteristics within these specific parts to be analysed in detail. By positioning the mass spectrometer (MS) capillary at selected locations and performing experiments with varying capillary positions, profiles are obtained for each individual compartment. The experimental results and data used in the study are derived from the results of these specific measurements and details can be found in the referenced report. Considering the scope of the current study, which focuses on the analysis of up to two layers of convolution, this investigation focuses on two key compartments: (1) the RTD of the MS capillary, which serves as the first layer, and (2) the RTD of the first pipe in the overall system, which represents the second layer.

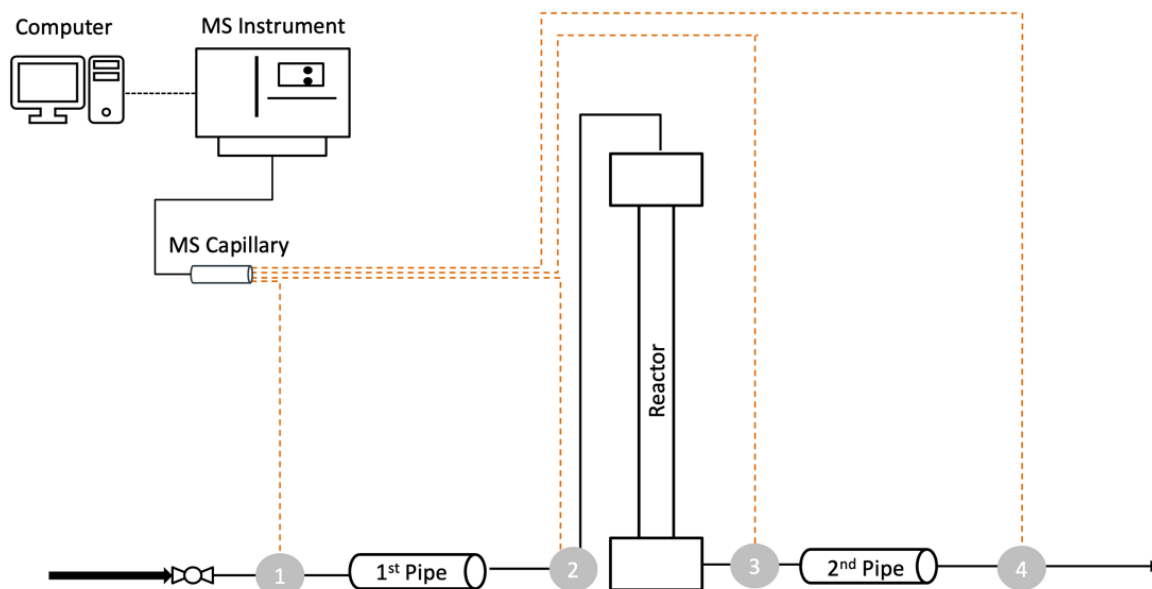


Figure 3. 1: The schematic description of the experimental set-up from initial position (1) to final measurement position (4) with measuring instrument MS.

Figure 3.2 illustrates the temporal response of positions 1 and 2 to a step input, plotted as normalised concentration profiles over time. The graph indicates a considerable delay in the response at position 2, due to the presence of an additional compartment in the system. Despite both measurements being taken at the same flow rate, this delay is evident because

of the additional time taken for the tracer to reach its maximum concentration at position 2. Nevertheless, to have a comprehensive view of the behaviour of the system, positional experiments are carried out at three different flow rates, thus, the concentration profiles are analysed under different conditions. The individual profiles of the datasets are presented in Figure A.1.

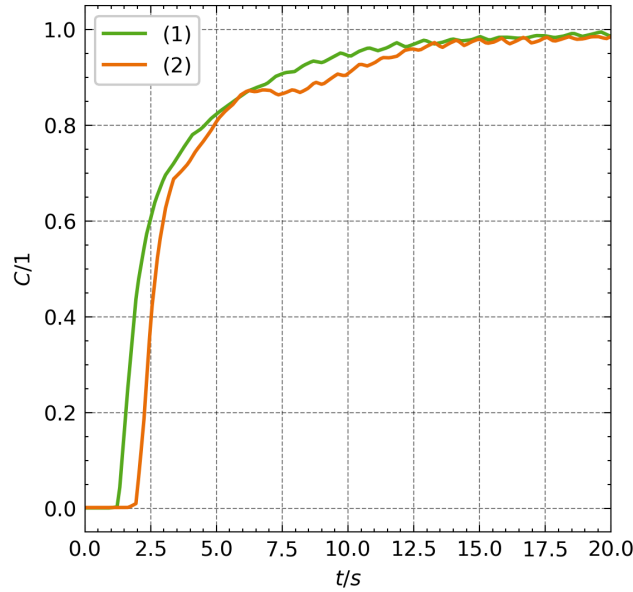


Figure 3. 2: Development of the normalised concentration profile of the tracer at different locations (1) and (2).

3.2 Experimental training data evaluation

The recorded experimental data is subjected to a pre-processing phase to ensure that the proper temporal domain is captured. The process includes a temporal adjustment of the data sets, eliminating excess periods in order to focus on the targeted experimental period. Moreover, the experimental data include both the step-up and step-down responses of the tracer within the experimental setup, capturing the behaviour of the tracer from its introduction to its complete evacuation from the system. Therefore, the convolutional neural network implementation is designed to handle this two-phase response. The model architecture incorporates separate convolution operations in terms of the presence of the step-up and step-down functions by varying the input. Each element is specifically matched with its corresponding response profile, allowing for an accurate representation of system dynamics while preventing potential asynchronies between input and output of the model. This implementation supports the fact that CNN are not only able to identify a specific response, but also capture the underlying RTD, enabling them to predict the output for any given input.

3.3 Theoretical training data generation

The theoretical training data set is generated by individual implementation of each corresponding compartment model (Section 2.4). Laminar flow and dispersion models are

used as basic single compartment representations while Adler-Hovorka, Cholette and Cloutier and Unified Time Delay models are used for more complex scenarios involving multiple compartments as advanced models. The selected models are purposefully determined to cover a variety of compartmental mechanics and system behaviours. Thus, allowing for a comprehensive examination and linking computational results with experimental configurations in future studies. The implementation done in this work involves programming the mathematical formulations of the models into Python modules.

In regards to the laminar flow model, a flux-flux setup is employed, utilising Equation (2.18) with a residence time of 5 seconds. The choice of flux input and output is motivated by the necessity to simulate data comparable to experimental setups. The methodology allows a more direct comparison between model predictions and actual experimental data, thereby enhancing the practical applicability and validation of the model. The implementation of the dispersion model is conducted according to the Equation (2.15). The model is investigated via varying Bo numbers at 1, 10, and 100 while representing different data sets for individual training and maintaining $\tau = 5$ s.

The Adler and Hovorka model employ an analytical approach using the transfer function, as shown in Equation (2.24). The inverse Laplace transform via SymPy in Python derives the RTD function from the transfer function, accordingly the relation in Equation (2.22). The related part of the code is provided in Appendix B. Moreover, the exchange flow rate, is assumed to be 0.1, which is 10% of the actual flow rate. The residence time of the compartment of the PFR in the model is arbitrarily chosen as 2 seconds, while the residence time of the STR with stagnant volume is varied as 1 and 3 seconds.

Since the Equation (2.23) is used to implement the Cholette and Cloutier model contains the Dirac delta function, the E -function is convolved with a step function using the Wolfram computer algebra system instead of the Python environment. [36] A direct implementation of the Dirac delta function is impractical due to limitations in the Python libraries, as the Dirac function cannot be explicitly defined in a numerical framework due to its infinite high order and infinite narrow width in Python. [8]. Nevertheless, the approximation preserves the integrity of the model and ensures that the critical effects of dead volume and short-circuit flow are effectively captured in the RTD analysis. The bypass flow is established at 20% of the total flow, while the dead volume is analysed at β values of 0.1, 0.3, and 0.9, maintaining $\tau = 5$ s.

The Unified time delay model is implemented based on Equation (2.24), which defines $G(t)$. While the transfer function requires the inverse Laplace transform, the complexity of the exponential term, which varies with 2 and 5 well-mixed cells, limits its calculation in Python.

Consequently, the analytical implementation and calculation of the $E(t)$ function is performed using the Wolfram computer algebra system [36]. The model assumes a mean residence time of between 1 and 3 seconds, including an additional 1 second delay due to lateral flow.

The individual normalised $E(t)$ function of each model, except for Cholette and Cloutier, is convolved with an ideal step-up function, $C_{in}(t)$, which has an instantaneous rise at $t = 5$ s. The convolution technique from Equation (2.9) is numerically implemented in Python via NumPy's `convolve` method to generate the overall RTD profiles. The convolution operation effectively distributes the step input over time according to the characteristics of each $E(t)$ function, resulting in concentration profiles that describe the evolution of the system according to the corresponding literature model. The time length of the output profiles, t_{out} , is determined by the specific evolution of each profile. The duration is adjusted accordingly as the RTD of each model exhibits a unique response. Therefore, to fully capture the complete behaviour of each specific RTD model, the time scale for each output profile is tailored to represent its distinct temporal characteristics. Furthermore, the generated synthetic datasets do not include additional interferences in investigating the individual theoretical compartment models. Once the data sets are generated, they are introduced into the developed CNN model to assess its ability to detect and identify the inherent residence time distribution.

3.4 Neural network construction

In this study, a neural network RTD (nRTD) model is developed based on the convolution technique and deep learning is implemented using the CNN. The implementation of this approach is enabled by the conceptual similarities between CNN and RTD analysis, particularly in their use of convolutional methodology. Both CNNs and RTD analysis employ element-wise multiplication and convolution operations to transform input data into meaningful outputs. In RTD analysis, the input concentration is convolved with an RTD function to predict the output concentration, as shown in Equation (2.13), while, in CNNs, the input data is convolved with weight functions to generate predictions, as indicated in Equation (2.27). The key parallel lies in the correspondence between the weight functions in CNN and the E -function in RTD analysis. This analogy forms the basis of the nRTD model, where the convolution operation in CNN is adapted to process RTD functions. Moreover, according to Equation (2.27), CNN includes a bias term. However, since the RTD convolution formula, Equation (2.13), does not include any additive term, the bias term is kept out of the implementation of the nRTD model. Consequently, the convolution operation in the neural network model is consistent with the RTD analysis; hence the model is structured so that the corresponding learned weights represent the RTD function. Equation (3.1) represents the mathematical methodology of the proposed model for convolution layer i and theoretical model of j as it takes

the input function, $C_{i,j}$, and convolves it with the RTD function, $E_{i,j}$, to produce the predicted system output concentration, $\hat{C}_{i,j}$.

$$\hat{C}_{i,j} = C_{i,j} \otimes E_{i,j} \quad (3.1)$$

The nRTD is a PyTorch neural network module designed for RTD analysis by applying convolutional layers within the framework of compartmental models and is designed to be suitable for evaluating temporal inputs and corresponding responses, RTD profiles. The modelling of nRTD is theoretically separated into two classes, as RTD_{Net} classifies the architecture of the neural network, while RTD_{Fit} is responsible for managing both forward and backward propagation with training parameters.

The RTD_{Net} submodel defines the structure and configuration of the convolutional layers. These layers are responsible for capturing temporal patterns using the provided network. In this particular implementation, a one-dimensional (1D) convolutional layer is implemented. After the successful implementation of the first layer, multiple 1D convolutional layers are executed using the sequential workflow approach. Each convolutional layer operates on an input data, the raw data fed into the neural network, with a predicted kernel. The convolution process involves sliding the kernel over the input data and computing the element-wise product between the kernel and the overlapping input elements [31-33]. This process is repeatable for each kernel, with each kernel corresponding to one convolutional layer. Moreover, specifically, a convolution layer convolves the input data C_{in} with the predicted kernel, E , resulting in the output C_{out} with discrete point of $n_{i,i}$, $n_{e,i}$, and $n_{o,i}$. The sequential workflow of the network architecture is illustrated in Figure 3.3. The Figure 3.3 shows that each layer is interconnected, including the input, convolution, and output layers.

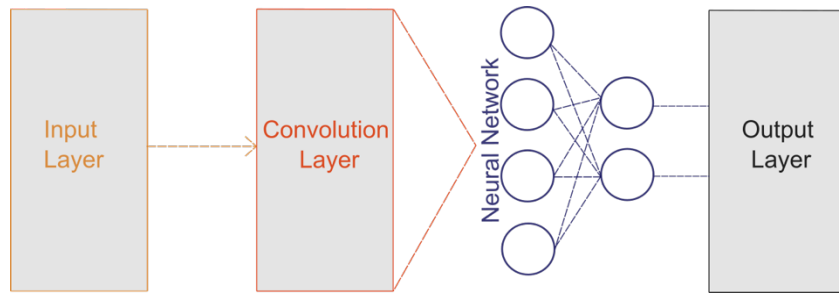


Figure 3. 3: Network architecture of nRTD model including input, convolution, and output layer.

The convolution operation, where a kernel traverses the input data, can reduce the number of points in the predicted output depending on the convolution mode applied [37]. Valid convolution, which only processes regions where the kernel completely overlaps the input, results in a reduced output size without extending the input data beyond its boundaries. This approach results in a loss of points at the edges, as the filter excludes regions that it cannot

fully cover. In contrast, full convolution, which includes padding around the input, ensures that the kernel can slide over all possible regions, including the edges [4]. This produces a larger output by considering full overlaps at the boundaries and extending the input data. For RTD profiles, preserving information over the entire profile, including the boundaries, is critical, as the entire profile is important for analysis. Therefore, full convolution is chosen for this implementation. By capturing information over the entire input, including padded edges, full convolution ensures that RTD analysis is performed without loss of points, maintaining the integrity of the entire profile and preserving boundary features for accurate analysis.

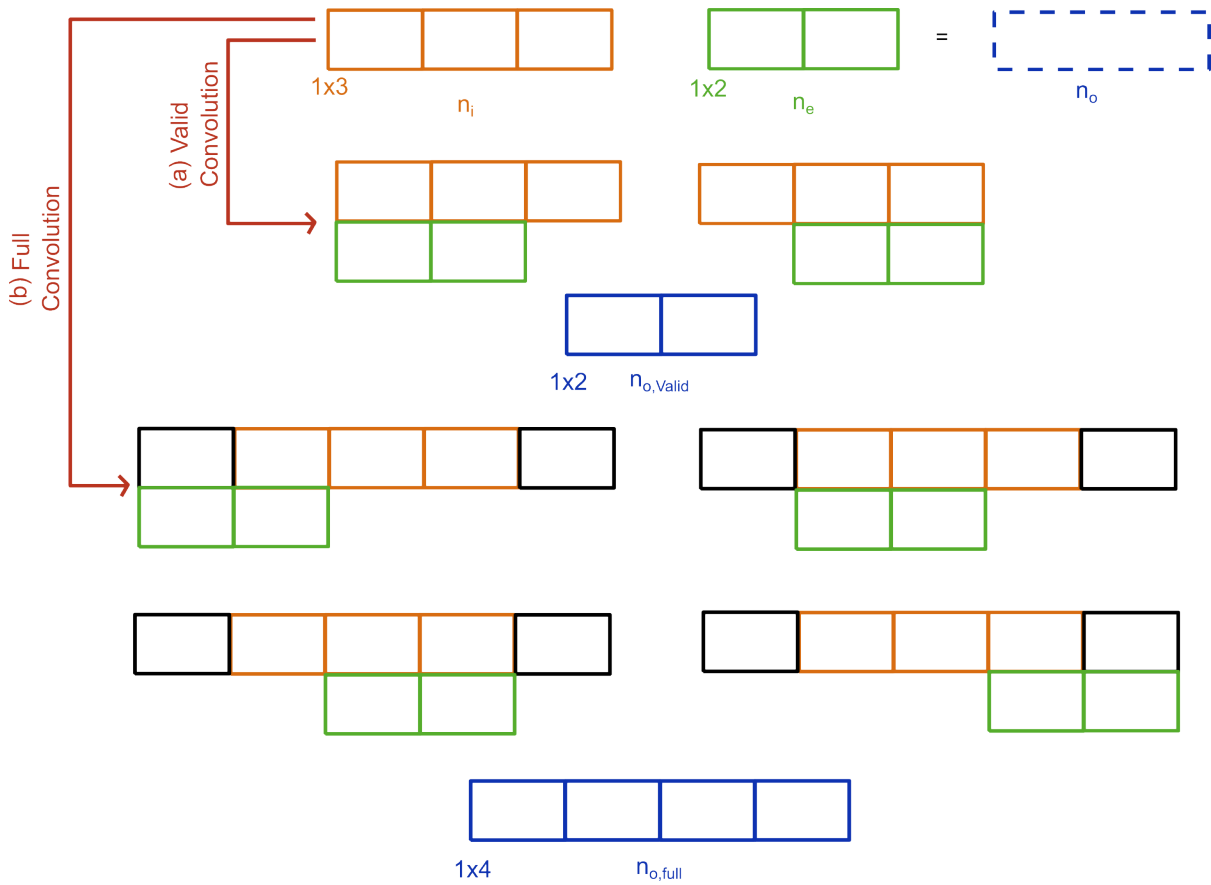


Figure 3. 4: Comparison of valid and full convolution operations. The input (orange) is convolved with a kernel (green), showing the effect of padding (black) on the output (blue) for both convolution modes.

Figure 3.4 illustrates these concepts by showing how a 1×2 filter applied to a 1×3 input produces different outputs in valid and full convolution modes with 1×1 padding. In valid convolution, route (a), no padding is applied to the input, meaning the kernel slides over the input data without extending beyond its boundaries [4]. Therefore, the output is smaller, 1×2, than the input dimensions, as edge regions are excluded. Conversely, with full convolution, route (b), padding is added around the input, shown in black. Thus, the filter can slide over all possible regions, including the edges, and the padding allows the 1×2 filter to produce a larger output by covering the extended input. This results in an output larger than the input because

the padded regions allow the filter to compute values even at the boundaries [37]. In order to perform edge padding on the input to prevent shrinkage of the output, zero padding and replicate padding can be considered. Zero padding involves adding zeros around the input data, as shown in Figure 3.5 (a), which can result in a loss of edge information. In contrast, replicate padding increases the input size by extending its edges with duplicate values, as illustrated in Figure 3.5 (b), thus maintaining the integrity of the original data. While both methods prevent output shrinkage, replicate padding is preferred in this study because it preserves edge information, which is critical for RTD analysis[31-33]. Whereas using zero padding can lead to incorrect convolution at the padded edges, potentially resulting in inaccurate analyses.

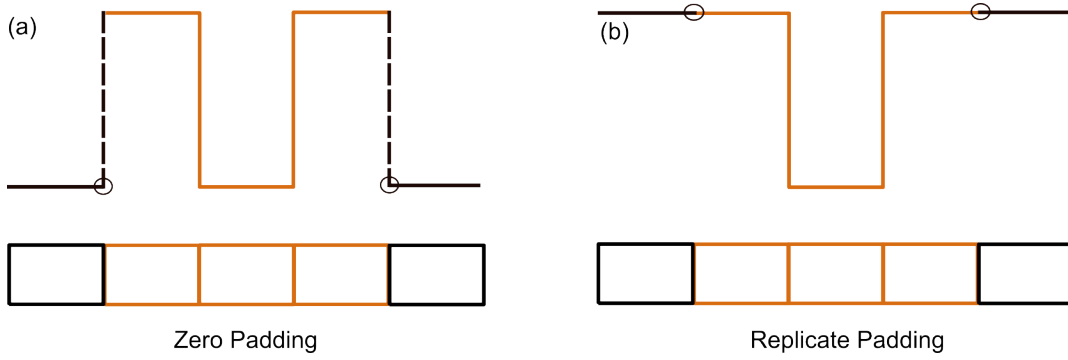


Figure 3. 5: Illustration of padding techniques on input data: (a) zero padding and (b) replication padding.

Consequently, the full convolution phenomena with the replicate padding mode that is implemented in this study is mathematically expressed as Equation (3.2). In this equation, the length of the total number of points in the processed signal, $n_{o,i}$, for the i^{th} convolution layer, is equal to the sum of the length of the input signal, $n_{i,i}$, and the kernel size, $n_{e,i}$, minus one [4].

$$n_{o,i} = n_{i,i} + n_{e,i} - 1 \quad (3.2)$$

Since it is critical to ensure that each kernel element convolves with corresponding input elements, computes their product, and sums for output, the arithmetic of convolution in time is defined as Equation 3.3 [13]. While $\Delta t_{o,i}$ and $\Delta t_{i,i}$ define the time scale of output, and input, $\Delta t_{e,i}$ represents the time scale of the convolutional kernel and thus of RTD function.

$$\Delta t_{o,i} = \Delta t_{e,i} + \Delta t_{i,i} \quad (3.3)$$

Since every RTD profile is distinct in terms of the evolution of the curve over time, it is essential to define the kernel time, $\Delta t_{e,i}$, with care to enable the kernel to extract the RTD within the specified time frame effectively.

Therefore, $\Delta t_{e,i}$ is explicitly defined for each investigation of a profile according to the time required from the moment the tracer is injected till the profile reaches the steady-state value of the input function. Since the convolution needs both the presence of input and kernel for convolving each to be able to get the pattern in concentration profiles, the time of the input data, $\Delta t_{i,i}$, is implemented in a similar time domain. Thus, the total time is determined according to the compromise between the time required for the kernel and the input. Also, since consistency in the full convolution operation in terms of both the kernel and the input should be maintained, Equation 3.4 must be maintained within the model across these different scenarios in each individual implementation. Thus, the convolution process ensures the accurate capturing of the characteristics of each RTD profile in each [13].

$$\frac{n_{e,i}}{\Delta t_{e,i}} = \frac{n_{i,i}}{\Delta t_{i,i}} \quad (3.4)$$

Moreover, due to the variation in the development of the RTD profiles, Equation (3.2) is combined with Equations (3.3) and (3.4) to calculate the parameters of the kernel and the input in the implementation while ensuring proper aligned convolution between them. The $n_{o,i}$ is determined based on the findings presented in Section (4.1). Thus, Equation (3.5) is obtained for the calculation of the input signal.

$$n_{i,i} = (n_{o,i} + 1) \left(\frac{1}{1 + \frac{\Delta t_{e,i}}{\Delta t_{i,i}}} \right) \quad (3.5)$$

In the model development process, calculations are implemented individually for each specific investigation, while the underlying mathematical framework and equations remain consistent across all studies, providing a consistent approach to modelling while allowing adaptability to varying conditions. The model parameters for the laminar flow model are listed in Table 3.1. Further calculations, including parameters for additional compartmental models implemented, are provided in the Appendix C.

Table 3. 1: Discretisation and temporal parameters for the nRTD model applied to the study of the laminar flow model.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	101	100	101 s	51 s	50 s

The kernel is initially assigned random signal values, which serve as the starting points for the convolution process and further and each system output is the result of the convolution of the currently predicted kernel, RTD, and input data.

The RTD_{Fit} class is developed to perform the processes of fitting and assessing the model. It provides a layer of functionality that extends the structured RTD_{Net} module and includes functionality for model training. Therefore, RTD_{Fit} encapsulates the four key steps in the nRTD model: forward pass (performing convolution calculations), loss calculation, backward pass (computing gradients), and parameter optimization. The training process is carried out through an iterative optimisation that spans a series of epochs. An epoch refers to a complete pass through a training data set during the training process of the model where convolution occurs. In the context of this study, the training data set may consist of either synthetically generated data or experimental data, as described in Section (3.2) and Section (3.3). Subsequent to each training step in a single epoch number, backpropagation begins, and the loss function computes the error based on the current model parameters. Hence, Mean Squared Error (MSE) is utilized to calculate difference between the predicted values generated by the model, $\hat{x}_i$, and the actual target values in the dataset, x_i . Therefore, Equation (3.6) indicates the mathematically expression of the loss calculation where the summation is made across K samples in the dataset [38].

$$\text{MSE} = \frac{1}{K} \sum_{i=1}^K (x_i - \hat{x}_i)^2 \quad (3.6)$$

At the end of each epoch, the training loss is systematically logged, providing a comprehensive record of the model's performance throughout the training process. This systematic logging captures the trend of loss over time, enabling further analysis. Once the backpropagation process is complete, the model requires an optimiser to update its parameters further based on the resulting training loss. The ability of the model to recognise patterns and relationships within the RTD data depends on the iterative learning process as the model progresses through successive epochs, it gradually adjusts its internal representations. Thus, the parameters of the model are progressively improved to increase its predictive accuracy. In this implementation, the optimiser reduces the error between predicted and actual values by adjusting the weights to minimise the loss function [38]. Backpropagation and optimisation work in cooperation, where backpropagation calculates the appropriate adjustments to the weights, and the optimiser applies the adjustments, converging on an optimal set of parameters that minimises the error. Therefore, the Adam optimiser is defined for this specific application. The testing workflow begins after the iterative training process concludes. The predicted output of

the model is evaluated using data that is not seen during training to assess the effectiveness of the model. For some specific cases investigated in this study, the same data set which is to investigate the predictive accuracy of the model is used for both training and testing the model, as well as for calculating the loss values, referred to as train/loss and test/loss in this study. The underlying reason for the approach is to emphasise the availability of training data over the necessity of a separate, unseen dataset. Thus, the data set is not split, resulting in identical train/loss and test/loss values for these particular model instances. Moreover, consistency in evaluating the results is maintained using the same error calculation methodology, MSE, throughout the process. Consistency in evaluating the results is maintained using the same MSE calculation at the end of each training cycle and the testing where after the model has finished training on all epochs. Thus, the comparability of the error is acquired.

The reproducibility of the model's output is essential to ensure that the behaviour of the model is consistent throughout different runs and that the outcomes within the model can be accurately compared. In order to ensure the repeatability of the model results, the deterministic mode is implemented in the model to ensure the reproducibility of the model training and evaluation processes. The deterministic mode provides consistency in data handling by eliminating randomness in the selection and use of data batches during both training and testing. This approach ensures that the same sequence of data samples is used in each epoch, enabling reproducible results.

The nRTD model incorporates three hyperparameters that significantly influence its predictive accuracy for RTD profiles. Two of these hyperparameters, namely the learning rate and the number of epochs, are explicitly defined and directly impact the model's ability to capture underlying trends in the RTD data. The learning rate is a hyperparameter that controls the magnitude of weight updates during training. It determines the size of the steps taken towards a minimum of the loss function during optimisation [31-33,37]. An optimal learning rate should be low enough for accurate convergence yet high enough for reasonable training time. Smaller rates require a high number of epochs, and the process may get stuck at some point so that no improvement can be observed. Larger rates, on the other hand, can cause fluctuations around the optimal solution with fast model convergence [37]. One of the conventional learning rate strategies is to define a constant learning rate as the basic method; model training with a fixed learning rate strategy is used. Conversely, the adaptive learning rate method for training involves changing the learning rate by a predefined value if no improvement in accuracy is observed after a few epochs. Therefore, fine-tuning the learning rate is essential for practical training, and techniques such as learning rate scheduling can help achieve this balance, improving both speed and performance. In this study, a learning rate scheduler is used by

defining factor and patience parameters. The factor defines the degree to which the learning rate is decreased if the performance of the model becomes stagnant, while the patience parameter specifies the number of epochs to have waited until the learning rate is reduced if there is no performance improvement. Thus, the learning rate is adjusted according to a dynamic schedule, reducing it by a factor of $\lambda = 0.5$ after eighty epochs without a relative change of 1×10^{-4} in training error [39]. The number of epochs hyperparameter, representing complete passes through the training dataset, is critical in determining the extent of the model's training on the data, thereby influencing its ability to learn patterns. The third hyperparameter of the nRTD model is the level of discretisation of the output dataset, $n_{o,i}$. Although this parameter is not directly involved in the learning process, it indirectly affects the performance of the model by determining the resolution at which the RTD profiles are represented, thus affecting the size of the characteristics that the model can learn and predict. The investigation of the effect of the hyperparameters on the model performance is carried out in Section (4.1).

3.5 Multilayer implementation

The implementation of the second convolutional layer is built after successfully modelling the first layer, ensuring that it is built on a solid foundation. The sequential approach ensures that the design and parameters of the second layer are designed based on the established performance of the previous layer to analyse RTD in multi-compartment systems. Also, it supports the model in capturing the sequential dynamics of multiple compartments and their individual and collective contributions to the overall RTD. Since each compartment demonstrates a unique response profile, specific kernels are required for each, thus requiring individual convolutional layers customised for the responses. The first convolutional layer processes the initial input data and produces an output describing the response of the first compartment. The resulting output encapsulates the dynamic behaviour of the first stage, transformed by the learned weights of this convolutional layer. The second convolutional layer then takes the processed output as its input, thereby modelling the behaviour of the next compartment in the sequence. Thus, the second layer effectively builds on the processed information of the first layer, and this coupling ensures that the sequential dependency inherent in the system is maintained by capturing the cumulative effects of each compartment on the overall system dynamics. Therefore, the layers are not independent and do not operate simultaneously; instead, they are connected in a cascading arrangement, with the output of one layer feeding into the next. Figure 3.6 provides a schematic illustration of the cascading structure of the nRTD model, focusing on the layered hierarchy. It represents the interconnection between layers, indicating the processing of inputs through convolutional operations with respective kernels. The Figure 3.6 illustrates the relationship between inputs

and kernels across layers without detailing the particular scale differences, focusing on the conceptual framework of the model.

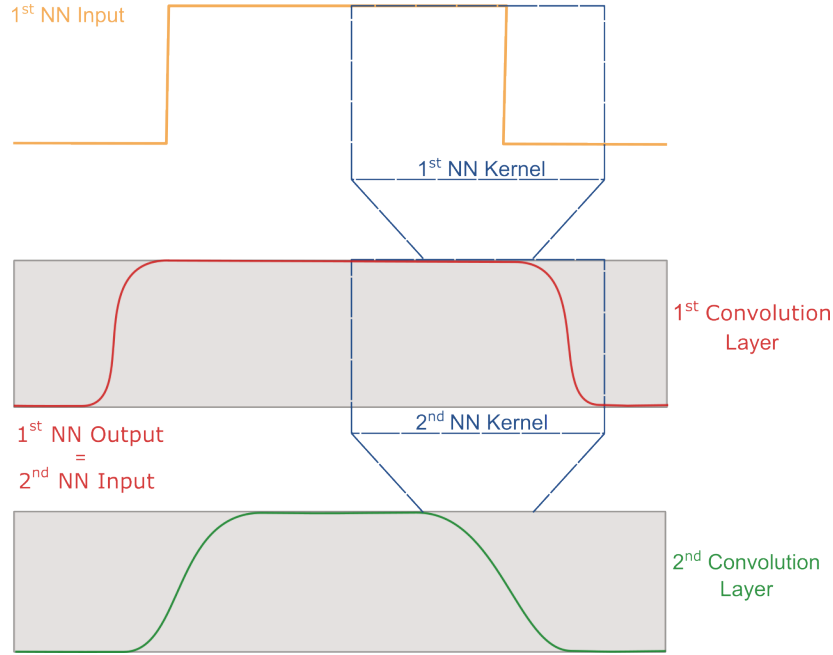


Figure 3. 6: Scheme of the cascading nRTD model with the layered convolutional operations.

The design of the second layer follows the same principles as the first layer, implementing the mathematical concepts described in Equations (3.2) through (3.5). However, the structured second layer follows the mathematical principles of convolution operations, where successive convolutions build on prior results. Therefore, the degree of freedom in selecting independent parameters is reduced due to the dependence of the second layer on the output of the first layer, as shown Figure 3.7.

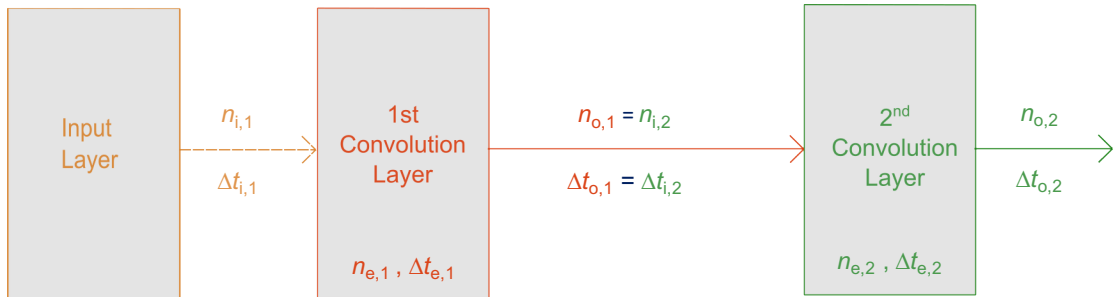


Figure 3. 7: Schematic illustration of parameter relation between network layers.

Due to the cascading design of the nRTD model, the RTD function of the second compartment, $E_2(t)$, inherits most of its parameters from the previous layer while only the time scale of the convolutional kernel, $\Delta t_{e,2}$, is chosen arbitrarily. Table 3.2 presents the model parameters for the two-layered nRTD model, combining a laminar flow model in the first layer

with an Adler-Hovorka model in the second layer. Further calculations, including parameters for additional compartmental models implemented, are provided in Appendix D.

Table 3. 2: Parameters of the 2nd layer convolution in the nRTD model: Laminar model convoluted with Adler-Hovorka compartment model.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$	$n_{o,2}$	$n_{i,2}$	$n_{e,2}$	$\Delta t_{o,2}$	$\Delta t_{i,2}$	$\Delta t_{e,2}$
Value	200	101	100	100 s	51 s	50 s	296	200	97	150 s	101 s	49 s

4 Discussion

In order to improve the predictive accuracy of the nRTD model, the hyperparameters, which are epochs number, the learning rate and the number of discretisation of the dataset, are examined before the training datasets are evaluated. The laminar flow model, one of the fundamental flow regimes in fluid dynamics, serves as the basis for hyperparameter studies and complementary investigations due to its relative simplicity. The model serves as an effective baseline while providing a straightforward framework to modify and extend to incorporate additional considerations without significantly complicating the analysis, making it easier to identify optimal configurations and validate results across different conditions.

4.1 Hyperparameter study

4.1.1 Learning rate variation

The investigation of the learning rate over the range 10^{-5} to 10^1 is used to understand the response of the deep learning-based model to hyperparameter variation.

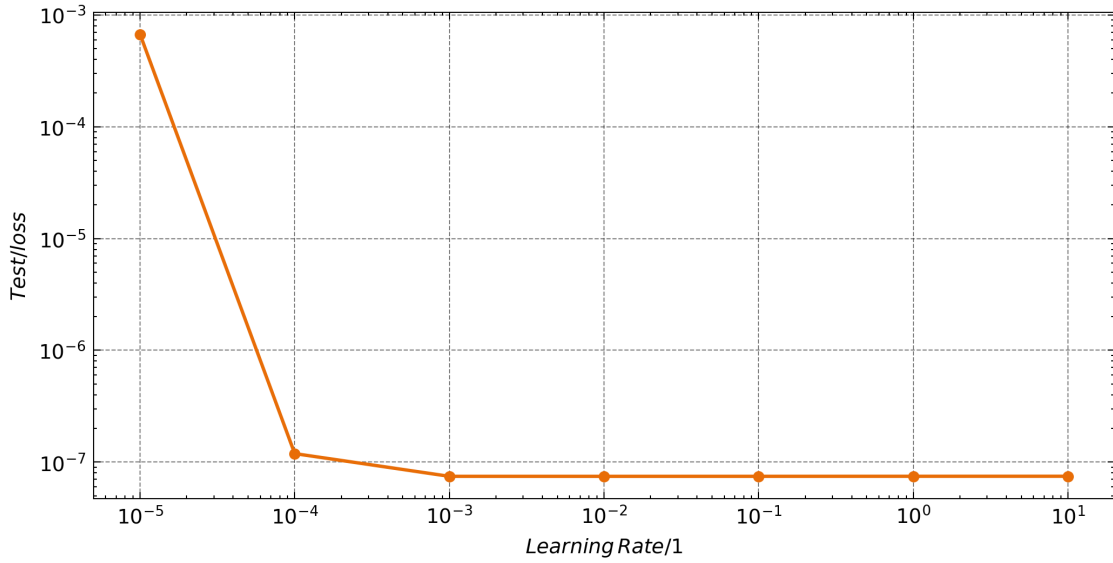


Figure 4. 1: The varying predictive accuracy of the model with different learning rates.

Figure 4.1 illustrates the test/loss values calculated at the end of the training process corresponding to different learning rates on a logarithmic scale. The test/loss represented in this section are evaluated at a specific number of epochs, as defined in Section (4.1.2). According to Figure 4.1, the lowest learning rate considered, 10^{-5} , performs as a limit, resulting in relatively high test/loss values and adversely affecting the performance of the model. The behaviour occurs due to the effect of the low learning rate, leading the model to update its parameters rather slowly, which prevents sufficient progress towards the global minimum of the loss function. Consequently, the model demands an overly long run time to reach a comparable test/loss value within a given range of epochs [37]. Similarly, a learning rate of

10^{-4} causes higher test/loss values at the given epoch, and thus it takes longer training times to achieve lower test/loss values. Figure 4.2 illustrates the phenomenon of slower convergence as train/loss versus epoch number. The learning rate of 10^{-5} requires the longest runtime to reach a comparable test/loss value, and the learning rate of 10^{-4} follows similarly. Therefore, the lower learning rates require a higher number of iterations to compensate for the initially high test/loss values, resulting in a more computationally intensive training procedure. Therefore, the optimum approach is to select a relatively higher learning rate, mainly because it tends to give better fittings although a slight decrease in the test/loss is seen between 1 and 10^{-3} . Furthermore, at higher learning rates, the model accuracy varies slightly and higher oscillatory peaks appears during training, as shown in Figure 4.2. It illustrates the loss values of the different learning rates during training, corresponding to each epoch number on the y-axis, on a logarithmic scale. At the highest examined parameter value of 10, the model's training profile exhibits significant instability, characterized by high peaks and a lack of smoothness. Consequently, this parameter is not considered as optimal performance. Even though the test/loss values of the model remain approximately the same between learning rate of 10^{-3} and 10, 10^{-2} shows the lowest test/loss value. Hence, the learning rate of the nRTD model is selected to indicate the complex interplay between learning rate dynamics and model performance.

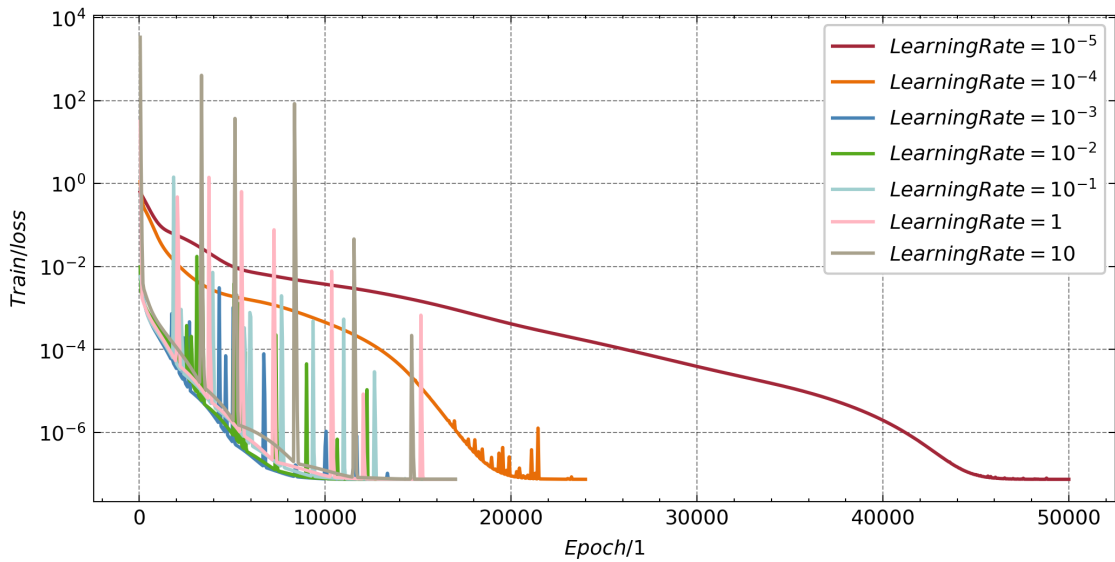


Figure 4. 2: Comparison of the training profiles for different learning rates.

4.1.2 Variation of Epoch Number

The influence of the number of epochs, a hyperparameter that indicates the total number of cycles in which the entire data set is iteratively passed through the neural network during training, is investigated [31-33,37]. The number of epochs ranges from 100 to 20,000, with

individual runs performed at different intervals, and the test/loss for laminar flow model is shown in Figure 4.3.

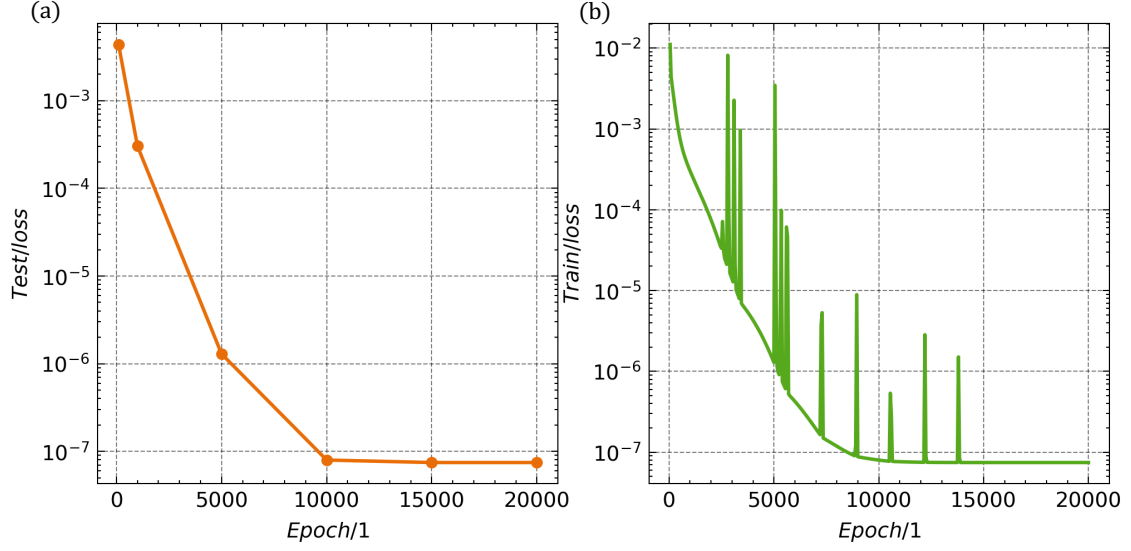


Figure 4. 3: The corresponding results of the model according to different epoch numbers (a) and corresponding training profile (b).

The required number of epochs is unique for each implementation since it is a function of the complexity and ability of the model to accurately represent the underlying patterns [37]. Consequently, the hyperparameter of the number of epoch, is tuned for each individual compartment model to reflect the characteristics of the underlying pattern. Furthermore, the increasing complexity of the investigated models results in an upward trend in the number of iterations required to reach a local minimum, which requires careful adjustment to ensure convergence. Thus, more accurate results on the performance of the model are obtained with each implementation. The specific hyperparameter values are reported in Appendix C. As shown in Figure 4.3.a, the model undergoes updates to its internal parameters at each epoch, resulting in an improvement in the fit so that higher numbers of epochs are initially correlated with lower test/loss values, indicating an improvement in the accuracy of the model. However, there must be a balance between computational resources and model performance since the model might need more iterations, and therefore, more computational resources are in demand in terms of processing power and memory. Additionally, as the number of epochs increases, the model is likely to converge to a local minimum in the loss landscape, and further iterations may not significantly alter the model. According to Figure 4.3.b, the local minimum has been reached at 15,000, epochs, and the further increment above that epoch number does not result in a lower test/loss value, conversely leading to the same model accuracy. Therefore, identifying 15,000 as the optimal number of epochs provides a compromise between achieving low test/loss, and computational efficiency while ensuring robustness to model performance without the excessive computational overhead.

4.1.3 Variation of Discretization

Since discretisation mainly involves in the convolution, it impacts the resolution of the continuous feature data. Consequently, the response performance of the nRTD model is affected and is therefore considered as a hyperparameter for this specific implementation. Figure 4.4 illustrates the varied output discretisation using the same epoch and learning rate parameters for each case. According to Figure 4.4, the resolution of the model output and test/loss value varies strongly. Therefore, a trade-off has to be found between relatively lower and higher time steps for computational resolution and model performance, 100 to 500, respectively.

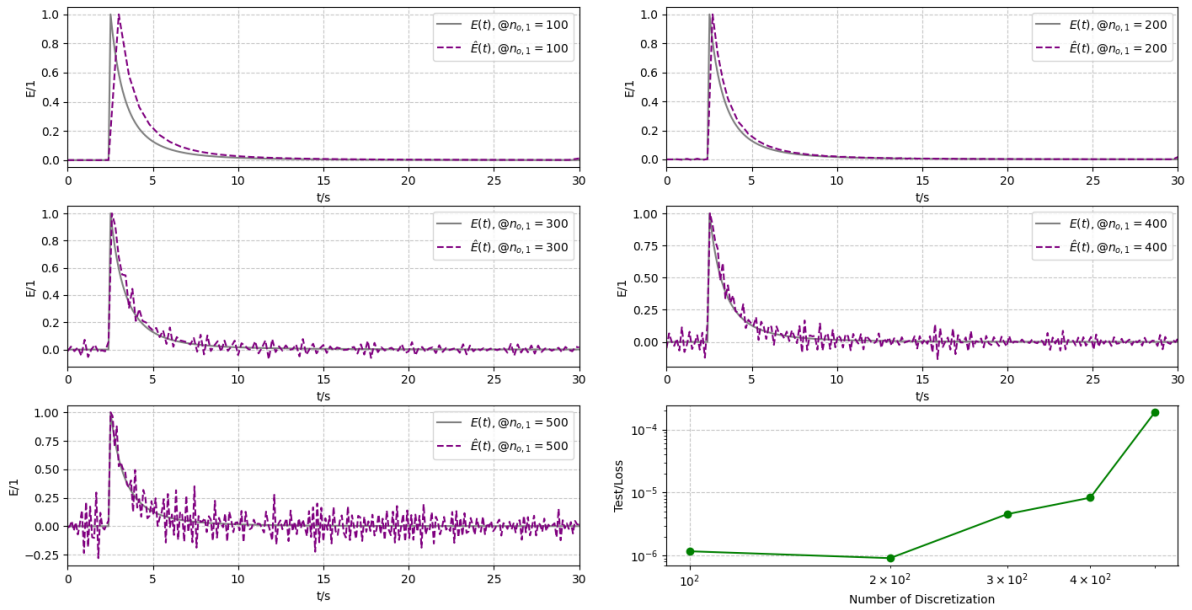


Figure 4. 4: nRTD model variation for disturbance investigation from 100 discretization to 500.

Moreover, the output of the model prediction is compared with the expected literature-based RTD profile, which is represented as the gray solid line in the graph. The input and the kernel size configuration are computed specifically for each case to match the size of the desired output, Equation (3.5). According to the model evaluation, it is found that the performance of the model is sensitive to changes in discretization and tuneable from a performance perspective.

At the lowest number of discretisation that are investigated, $n = 100$, the model exhibits significant deviations from the actual profile. The most significant deviation of the profile is observed for the region of the immediate response of the dynamic change in the beginning. With lower-resolution data, the model is able to achieve low loss values without fully capturing the underlying complexity, as it effectively represents a simplified version of the true

mechanism. The test/loss value is not significantly penalised by this over-simplification as MSE is based on point-wise differences between predicted and expected distributions [38]. The RTD profiles begin to approach the expected profiles as the discretization increases. This indicates that the model is able to capture both transient and steady state behaviour with increasing resolution starting from $n = 200$. Although the corresponding test/loss curve in Figure 4.4.d shows a directly proportional relationship with the increasing number of discretisation as a whole, at $n = 200$ an optimal behaviour with lower test/loss values at higher discretisation is observed. Initially, the test/loss value diminishes with increasing amount of discretisation from $n = 100$ to $n = 200$, as the increased level of resolution allows the model to improve its ability to capture the behaviour of the system by minimising the errors caused by rougher sampled data. At higher levels of discretisation, $n = 300, 400, 500$, the profiles have more data points in the same time interval, however, the presence of oscillatory behaviour impacts the projected profiles by generating deviations in the predicted RTD profiles compared to the smooth expected profiles. Thus, $E(t)$ becomes less interpretable. Whereas the overall shape of the profiles remains correct and allows the model to detect more detail, the oscillations result in an increase in the test/loss values. In terms of the observed oscillatory behaviour, the model may overcompensate for errors in its predictions. It can be characterised by a value that is too low, even below zero which is physically impossible for RTD function, being followed by a value that is too high, as the model attempts to balance the overall prediction. The results indicate that a higher number of data points leads to more complex optimisation problems in backpropagation, and thus less chance of identifying a global optimum. Therefore, at the intermediate level of discretization, $n = 200$, the resolution is adequate to acquire the primary features of the dynamics. The nRTD is able to learn and predict the RTD profiles with greater accuracy, reducing the test/loss value.

4.1.4 Effects of disturbances on the model performance

The effect of disturbance, which simulates measurement inaccuracy or noise within the dataset, is investigated to understand its impact on the model and to seek a potential solution approach to simulated realistic deviations. A series of data sets are generated based on the literature laminar flow model, and additional noise is introduced to the conventional convolved concentration profiles while the laminar flow function calculates the E -curve under the fixed time domains. In the implementation, the mean residence time is defined as $\tau = 5.0$ s and the discretisation are set to $n = 200$ points for all data sets. Additionally, randomisation is applied in the noise generation process while the predefined parameters remain constant. Therefore, training data sets are obtained by introducing noise into the RTD profiles. Consequently, 100 data sets with the same underlying laminar flow characteristics are obtained. The equivalent

mathematical representation is given by Equation (4.1) and Equation (4.2), where the noise, N is modelled as a normal distribution defined by its mean and standard deviation. The generated sample is shown as C' while the sample without disturbance is represented as C_{out} .

$$C' = C_{out} + N \quad (4.1)$$

$$N(\mu, \sigma) \quad (4.2)$$

A Gaussian distribution, where each noise value is sampled independently and centred around a mean of zero, is used to generate the noisy data. While maintaining a constant signal-to-noise ratio, SNR, the spread of this distribution is specified by a standard deviation proportional to the square root of the signal variance. The SNR is implemented based on the ratio between the signal, s , and N , defined in Equation (4.3) [40,41].

$$SNR = \frac{s^2}{N^2} \quad (4.3)$$

The implementation of the noise on the data sets is based on a variance-scaled noise generation method. For the evaluation of each data set, a variance scale is calculated as the product of a predefined noise level of 0.001 and the variance of the output concentration profile. The noise is subsequently generated from a normal distribution with zero mean and a standard deviation equal to the square root of this variance scale, whereby the approach results in a constant SNR across all datasets, regardless of signal amplitude. In specific, the SNR is set to 1000 using the variance-scaled noise generation method. Moreover, the technique ensures that the noise characteristics are proportional to the signal strength, providing a more realistic representation of measurement deviation in laminar flow systems.

The performance of the neural network is assessed by using an ideal laminar flow model, $C(t)$, as a test set. The training process involves a series of variations in the number of data sets with increasing samplings, 1, 20, 40, 60, 80 and 100. Figure 4.5.a illustrates the inverse dependence between the test/loss and the corresponding utilized data set for the training procedure. The improvement in performance as the size of the data set increases from 1 to 100 samples, with the loss decreasing in a non-linear manner, indicates that the model benefits from encountering a variety of laminar flow cases, thus capturing the patterns and relationships in the data. The resulting behaviour is consistent with the general principle in machine learning that larger datasets often lead to better model generalisation and reduced overfitting [31-33]. Although the model achieves relatively the lowest test loss on the largest data set, its performance on noisy data remains suboptimal. Specifically, the computed test loss/value for

the noisy data, even when using the largest training set, falls below the pure predicted. However, this behaviour is consistent with the concept of the aleatoric limit, which represents the model's appropriate response to inherent data noise while maintaining predictive accuracy without overfitting [43].

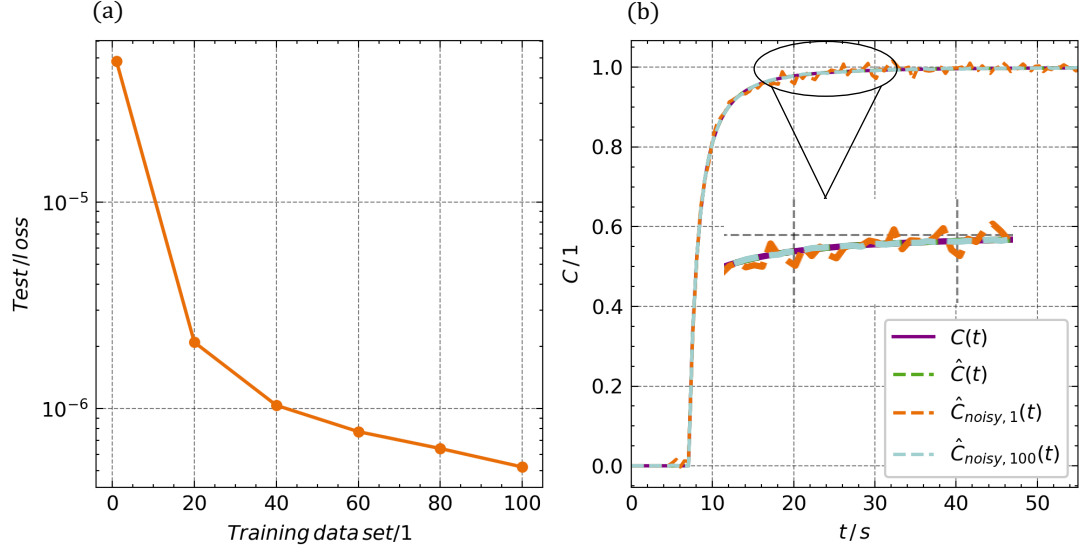


Figure 4. 5: Model performance with varying training dataset sizes (a) and corresponding concentration profiles (b).

Figure 4.5.b represents the normalised concentration profiles of the synthetic generated training data, $C(t)$, predicted output of the model without any noise in the training dataset $\hat{C}(t)$, predicted output of the model with noise on the noisy datasets 1 and 100, respectively $\hat{C}_{noisy,1}(t)$, $\hat{C}_{noisy,100}(t)$. For the smallest dataset utilized for the training, the ability of the neural network to learn and apply convolutional operations effectively is compromised by limited exposure to the underlying patterns and variations in the data. Consequently, the model struggles to distinguish between genuine signal characteristics and random fluctuations, resulting in challenges in determining the full depth of the concentration profile, especially $\hat{C}_{noisy,1}(t)$. Meanwhile, a significant improvement is observed for the largest dataset, where the concentration profile predicted from the model based on 100 datasets, $\hat{C}_{noisy,100}(t)$, closely aligns with the pure profile, highlighting that numerical test loss metrics reveal discrepancies. Also, the output profile shows strong agreement with baseline data, $\hat{C}(t)$, for larger training sets. By producing smoothed predictions, the model demonstrates its ability to avoid over-fitting to noise and instead learn and generalise to the actual underlying signal. Thus, increasing the dataset size improves the model's ability to generalise and handle noise.

Literature-based compartment model analyses

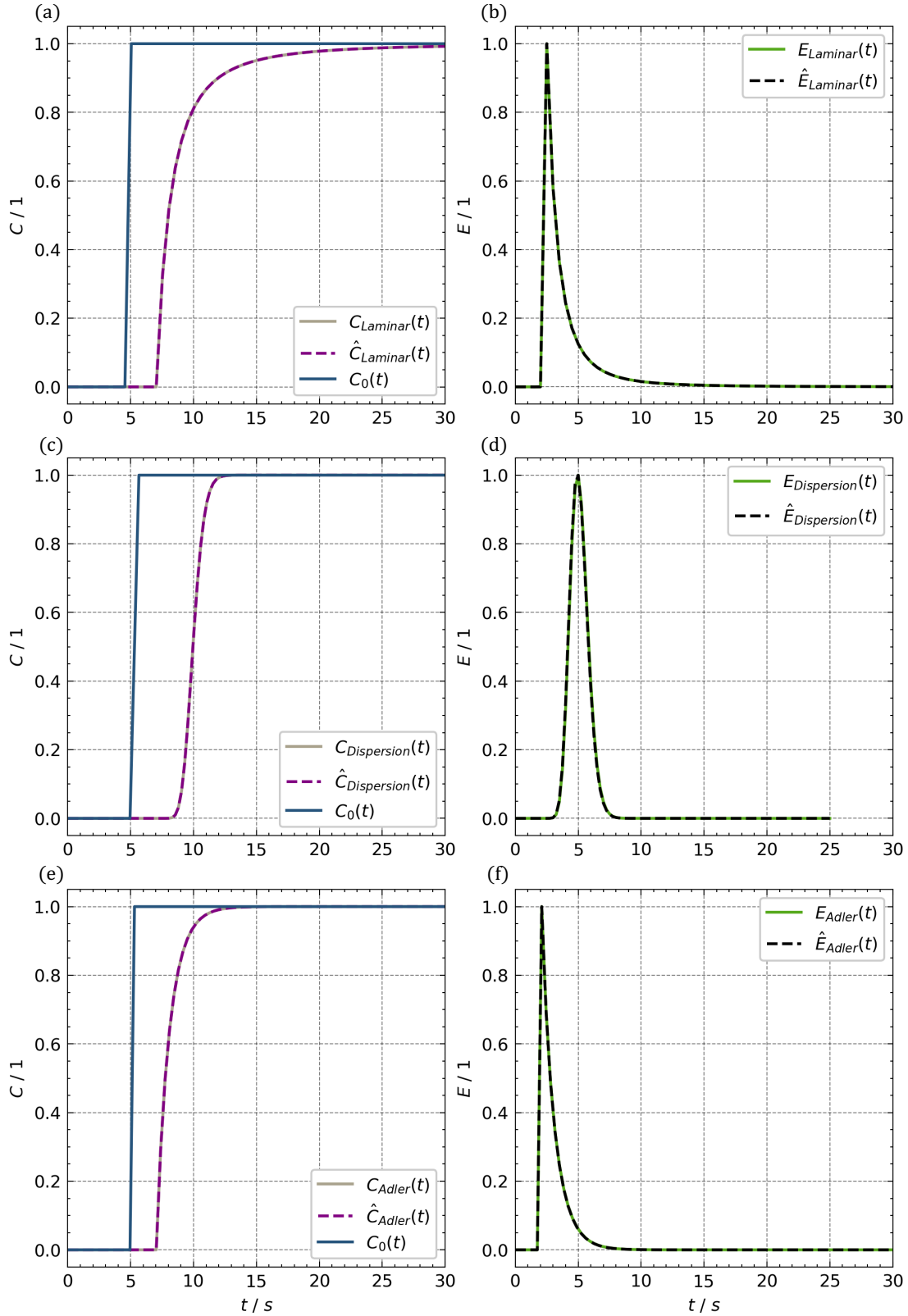
In order to assess the ability of the nRTD model to capture the feature maps associated with certain flow regimes and compartmental structures, multiple data sets are simulated based on well-established models from the literature. Single compartment, laminar flow and dispersion models are used to investigate the capability of the nRTD model. For more complex scenarios involving multiple compartments, Adler-Hovorka, Cholette and Unified Time Delay models are considered as advanced models. In addition, the synthetic data sets generated and utilized in this section do not include any additional disturbances. Afterwards, they are introduced into the developed CNN model to assess its ability to detect and identify the inherent complex residence time distributions and relations in the data. Individual data sets are used for both training and testing the model, as well as for calculating the loss values. The main reason for this approach is that as the number of training data sets increases, model performance does not improve further, since the training data is synthesized and lacks any potential perturbations. Furthermore, according to the hyperparameter study number, discretisation is utilised as 200 in each literature model as a basis to avoid any additional noise that can disturb the model response in the investigation. In order to meet the requirements of the convolution procedure, the output time of the model, t_{out} , is carefully defined. In addition, the time parameters are tailored to match the distinct dynamics of each kernel and input, ensuring that the convolution procedure captures the full behaviour of the system. Specifically, t_{out} , should be sufficiently long to encompass the entire duration of the kernel and to ensure that the input function presents for the convolution process, allowing sufficient time for the profiles to fully evolve under the combined influence of both the input function and the kernel. The detailed parameter settings for the data preparation and model implementation for each model can be found in Appendix C.

Figure 4.6 is an indication of the performance of the model for five types of models: Laminar, Dispersion and Adler and Hovorka, Cholette and Cloutier, and unified time delay, illustrated in rows 1, 2, 3, 4 and 5, respectively. There are two subplots in each row, and the left-hand column shows the concentration curves, while the right-hand column indicates the associated RTDs. For the normalised concentration profiles, the solid lines are the simulated profiles from the literature, $C(t)$, the dashed purple lines are the model predictions, $\hat{C}(t)$, and the blue lines are the input step function, $C_0(t)$. Furthermore, the right column contains the RTD functions, where the solid green lines correspond to the theoretical RTD functions, $E(t)$ and the dashed black lines represent the model predictions, $\hat{E}(t)$. The obtained results indicate a strong agreement between the model predictions and the simulated data for both the concentration profiles and the RTD functions at all points in time. Moreover, the functions which are the

outcome of the model, $\hat{E}(t)$, are indifferent from the corresponding simulated data, $E(t)$, appearing as the same lines in the plots. Thus, the model is an ideal predictor of expected system behaviour. Specifically, the concentration profiles derived from the model predictions, $\hat{C}(t)$, and simulated data, $C(t)$, consistently converge to the steady-state value of the step input signal. The capability of the model to accurately capture both the transient dynamics and the long-term equilibrium behaviour of the system is highlighted by this convergence. Thus, the overlap between the predicted and true profiles highlights the robustness of the model in accurately predicting both the concentration and RTD functions for different flow regimes.

In order to assess the predictive accuracy of the nRTD model on a convective flow dynamic, laminar flow model is investigated. Figure 4.6.a and Figure 4.6.b represents the RTD profile and function of the laminar flow model, respectively. According to Figure 4.6.a, the concentration profile for the laminar flow system has a characteristic delay in its dynamic response before reaching a steady state. The delay is observed as a gradual increase in the normalised concentration profile, although it is assumed that the tracer is introduced into the system at $t = 5\text{s}$ as an ideal step input. The delay represents the amount of time that is required for the step function to propagate through the flow system. As shown in Figure 2.3.b, Section (2.3.1), this behaviour is due to the parabolic velocity profile characteristic of laminar flow. Hence, different layers of fluid reach the measurement point at varying times, leading to a gradual transition in the concentration profile, shown in Figure 4.6.a. The concentration profiles, $C_{\text{Laminar}}(t)$ and $\hat{C}_{\text{Laminar}}(t)$ begins to rise at $t = 2.5\text{ s}$, corresponding to $t = \tau / 2$, where the initial sharp rise in concentration corresponds to fluid particles migrating along the fastest streamline, which is the centre line. It implies that even the fastest particles require time to reach the outlet. Therefore, for the laminar flow model, the neural network demonstrates the ability to capture the underlying flow dynamics. The profile of $\hat{C}_{\text{Laminar}}(t)$ accurately reflects the parabolic velocity distribution characteristic of laminar flow despite the absence of any formula in the model architecture. The neural network-generated profile exhibits the gradual rise and spread typical of laminar flow systems, reflecting the literature representation of tracer transport and mixing behaviour, including the capture of the delayed response and transition to a steady state. Importantly, the predicted, $\hat{E}_{\text{Laminar}}(t)$ and theoretical, $E_{\text{Laminar}}(t)$, functions are in perfect agreement, which directly translates into the agreement observed in the concentration profiles, thus, the underlying reason for the close match of the concentration profiles lies in the accurate prediction of the RTD function. According to Figure 4.6.b, the profiles, both $\hat{E}_{\text{Laminar}}(t)$ and $E_{\text{Laminar}}(t)$, are characterised by a narrow distribution with a sharp peak close to the mean residence time. Moreover, the sharp peak corresponding to the arrival

of fluid particles travelling along the fastest streamline is prominent in both predicted and theoretical profiles. The captured RTD profile demonstrates the ability of the deep learning-based model to accurately represent the fundamental characteristics of laminar flow dynamics [1].



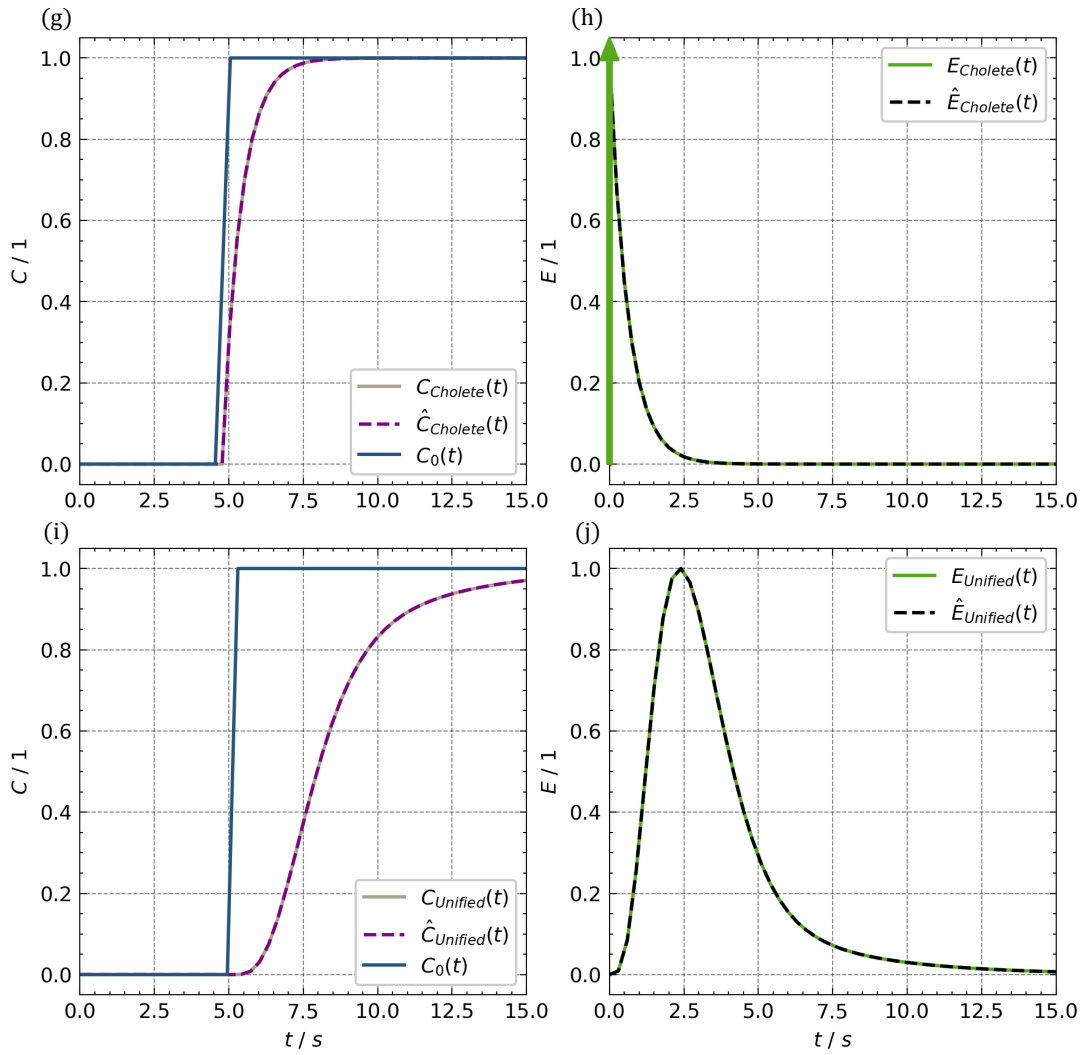


Figure 4. 6: Output of the literature based compartment models via nRTD model : (a-b) Laminar Flow, (c-d) Dispersion, (e-f) Adler and Hovorka, (g-h) Cholette and Cloutier, (i-j) unified time delay model.

The dispersion model is used to analyse the effect of the convection-to-dispersion ratio on the RTD and to evaluate the ability of the model to capture different intensities of axial back mixing. Bo of 1, 10 and 100 are investigated, representing a range from strong back mixing to near-ideal plug flow conditions. At $Bo=100$, as shown in Figure 4.6, in the second row, the axial dispersion is relatively small, resulting in limited back mixing. Furthermore, the tracer response at the outlet is relatively lagged as there is relatively insignificant dispersion to delay the fluid particles. Consequently, the concentration profiles, $C_{Dispersion}(t)$ and $\hat{C}_{Dispersion}(t)$ are narrow, almost equivalent to plug flow behaviour due to convection-dominated transport. Therefore, the sharpest profile in RTD functions is observed in the highest Bo assigned profile. Additionally, both $\hat{E}_{Dispersion}(t)$ and $E_{Dispersion}(t)$, demonstrate symmetrical E -curves at $Bo = 100$. The symmetry is driven by low axial dispersion, with fluid elements uniformly distributed around the mean residence time. The pattern of behaviour is consistent with the results reported by Güttel et al. and validates the implementation of the compartment model

[19]. Thereby, the ability of the model to accurately predict literature-based profiles for both $E(t)$ and concentration profiles further confirms the reliability of the CNN-based RTD analysis. The Appendix C, illustrates different profiles for different Bo values, demonstrating the sensitivity of the model to the degree of axial dispersion. Consistent with the literature, the concentration profiles are broader and spread around the $E(t)$ at lower Bo values, corresponding to increased axial dispersion [19]. The comprehensive investigation of Bo values (1, 10, 100) is effectively demonstrated and the ability of the model to accurately predict the variations underlines its ability to capture the flow dynamics and the transition from strong to weak back mixing cases.

In order to investigate the non-ideal flow patterns, which is particularly relevant as real flow systems often exhibit significant variability, Adler Hovorka model is examined [3,23]. The study investigates the variation on the residence time of the exchanged volume to understand the effect of a stagnant zone on the model. According to Figure 4.6.c, the normalised concentration transitions smoothly from 0 to 1, preceded by a noticeable delay. The lag is a feature of the PFR component in the model, which represents the time required for the fluid to pass through the plug flow section. The E-function of both the simulated data, $E_{\text{Adler}}(t)$, and the model output $\hat{E}_{\text{Adler}}(t)$ has two distinct characteristics, specifically a sharp increase at the beginning followed by a long tail at the end. The sharp increase is interpreted to be due to the presence of the PFR in the compartment model as a first step. Consequently, narrow concentration profiles are observed, indicating rapid removal of fluid particles from the system, corresponding to where fluid elements ideally maintain their sequence and experience minimal axial mixing. However, the model also includes back mixing through the STR compartment, which is reflected in the $E(t)$ as a long tail after the sharp rise, representing the gradual release of tracer molecules that have experienced longer residence times due to mixing in the STR domain. Therefore, the inclusion of an STR compartment addresses the non-ideal flow characteristics commonly observed in real reactors, resulting in a more gradual approach to the concentration profile to its steady-state value. Furthermore, the behaviour is investigated with the variation of $\tau_m = 3$ can be found in Appendix C. The longer tail of the concentration profile is observed in the aligned literature since a longer τ_m implies a longer time spent by the particles in the system [2]. Hence, the ability of the model to simultaneously capture both ideal plug flow characteristics and non-ideal mixing phenomena is demonstrated with the implementation of the Adler and Hovorka model, while supporting the variability and accuracy of the model.

The Cholette and Cloutier model is used to investigate the combined effects of dead volume and bypass flow on reactor modelling [3,22]. The RTD analysis of the Cholette and Cloutier model is implemented to evaluate the presence of dead volume, which does not exchange with

the active reactor volume, and the presence of bypass flow, which avoids mixing with the reactor contents. The actual implementation of the model includes a true Dirac function, which theoretically results in an infinitely high peak in the RTD function, as shown by the green peak in Figure 4.6.h. However, due to the inherent limitations of neural networks in representing infinite values, the CNN approximates this Dirac peak with a finite, but high value. Figure 4.6.g indicates a clear short-circuit profile, as revealed by the sharp rise in entry, even though input function the defined at $\tau = 5$ s. Corresponding to the fluid fraction that bypasses the active mixing zone and directly exits the system without interacting with the reactor contents, the $\hat{E}_{\text{Cholette}}(t)$ approximately captures this characteristic and exhibits a good agreement with the simulated results, shown in Figure 3.6.h. Furthermore, the $E(t)$ functions exhibit a sharp initial decay which is the peak at $t = 0$ in the kernel time, implying that part of the fluid leaves the system immediately after its entrance. Moreover, the effect of the dead zone is demonstrated in the tail after the initially pronounced rise in RTD function, which leads to a slower decay caused by fluid particles entrapped in the dead zone, and the resulting narrow $E(t)$ leads to a narrow concentration profile. Specifically, the model effectively catches the long-tailed behaviour of the $E(t)$ for increasing β values and thereby accurately represents the decreasing mixing performance without being penalised by the bypass behaviour.

Since the mixing on the lateral direction is present in real reactors and plays a critical role in understanding RTD analysis, the unified time delay model is investigated to demonstrate the capability of this phenomenon within the nRTD model with multiple zones. According to Figure 4.6.j, the behaviour corresponds to a low dispersion profile with similarities to PFR characteristics, especially for the highest number of cells investigated, $N = 5$ and $\tau = 3$ s. Therefore, for higher values of N , the model shows increased axial dispersion as the number of cells in the lateral compartment increases [1]. Moreover, the Figure 4.7 illustrates the effect of increasing the number of cells in the unified time delay model while the mean residence time, which is $\tau = 3$ s, is kept constant. Accordingly, the profiles of the $E(t)$ become more similar to the response of a PFR with the increasing number of cells. In particular, the residence time distribution becomes narrower, indicating that the fluid particles leave the system in a more localised time window. Furthermore, the model captures the formation of a comparable long tail in the $E(t)$ with increasing mean residence time. Thus, $\hat{E}_{\text{Unified}}(t)$ and $E_{\text{Unified}}(t)$ is the same function in all studied cases for the unified delay time model.

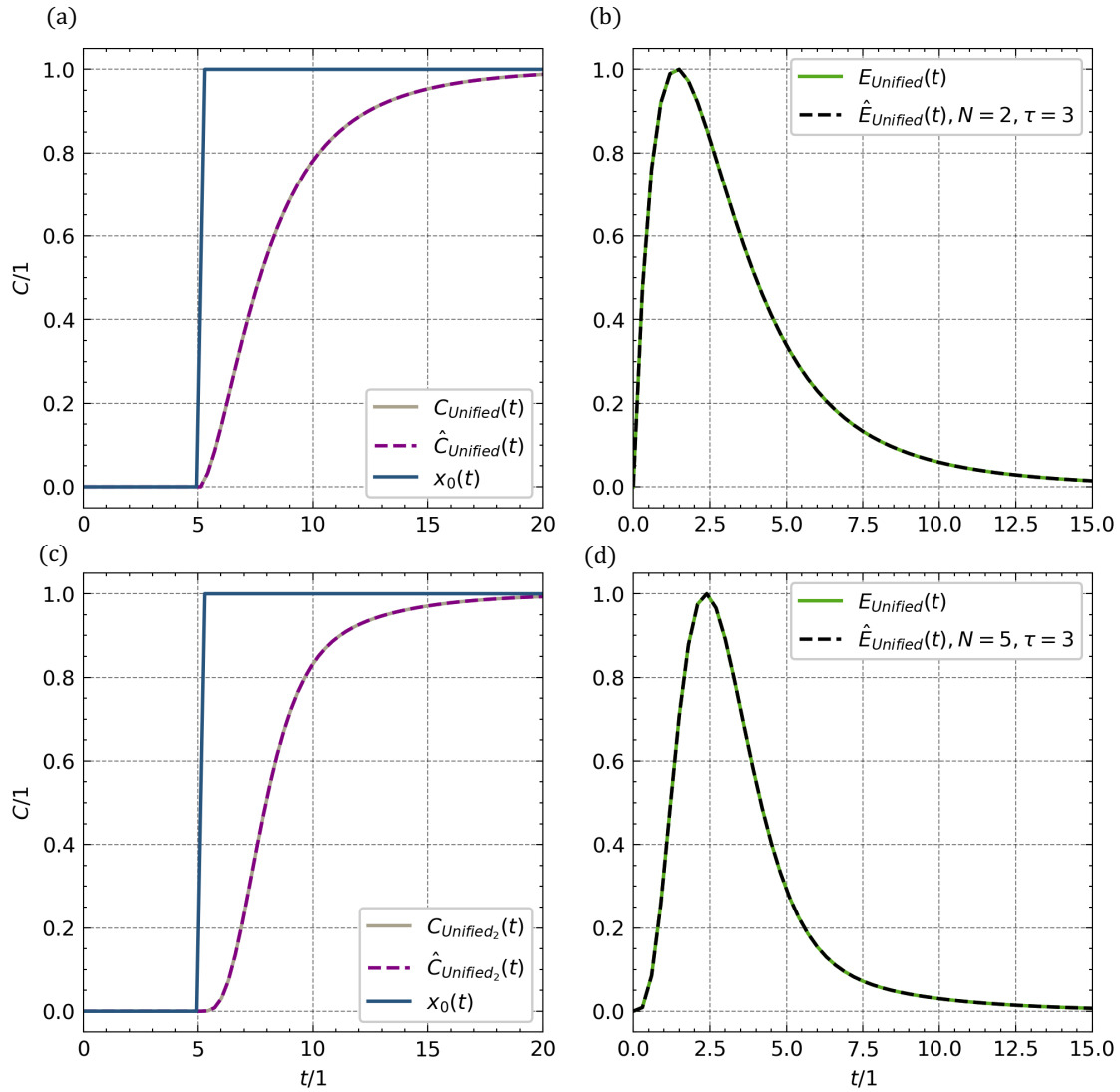


Figure 4. 7: nRTD model response to increasing number of cells in the unified time model: (a-b) The output of $N=2$ and $\tau=3s$, (c-d) The output of $N=5$ and $\tau=3s$

The validation of the first convolutional layer of the nRTD model using synthetic training data demonstrates its effectiveness in capturing a wide range of flow behaviours. As illustrated in Figure 4.6, the predictions of the model show strong agreement with the actual profiles derived from literature models. The nRTD model successfully captures convective behaviour in both RTD profiles and $E(t)$ functions for laminar flow, demonstrating its ability to accurately represent fundamental flow characteristics. The performance of the model extends beyond simple convection as it effectively handles axial mixing over a wide range of Bodenstein numbers. This versatility in predicting basic compartment properties is further complemented by its ability to model complex flow behaviours inherent in advanced compartment models. The nRTD demonstrates robustness in simulating phenomenal, short-circuits, dead volumes and stagnant zones, which are critical elements in complex flow systems. However, ensuring the robustness of the model to various literature models requires careful consideration of the convolution of

input and kernel, particularly in the transition from dynamic to steady-state profiles. In addition, the optimal number of epochs varies significantly between implementations, even within the same model with different parameters. This variability requires case-specific optimisation of the number of epochs to account for the unique RTD profiles generated by each model and parameter set. This comprehensive performance for simple and complex flow regimes, coupled with the need for tailored training approaches, highlights both the potential and the nuanced requirements of nRTD models in accurately predicting diverse flow behaviour.

4.2 Multilayer nRTD

4.2.1 Literature-based compartment model analyses for multilayer

In order to extend the compartmentalisation capabilities of the nRTD model, a multi-layer convolutional approach is implemented as described in Section (3.5). This improvement enables the representation of individual compartments within complex systems, allowing a detailed analysis of multi-compartment flow dynamics. In this study, the implementation is limited to a two-layer cascade configuration as a first step towards more complex architectures. The selection of individual models is based on a rational approach, where each layer provides a complementary representation of different RTD nuclei elements. Thus, a comprehensive assessment of the performance of the model in capturing complex residence time distributions across multiple interconnected compartments is studied. Additionally, the same data set is used for training and testing in both convolutional layers, without introducing additional external disturbances. Until the second layer, the system dynamics reflect the characteristics described in Section (4.2) for individual literature compartment models. However, at the second layer, the predicted concentration profile from the first layer, $\hat{C}_{1,j}(t)$, is used as input for convolution with the second layer rather than a step function, $C_0(t)$, thus, the characteristics result from the second layer differ from the individual model response. Consequently, the CNN predicted profile is derived from the methodology of Equation (4.4), utilising the neural network training process.

$$\hat{C}_{2,j}(t) = C_0(t) \otimes \hat{E}_{1,j}(t) \otimes \hat{E}_{2,j}(t) \quad (4.4)$$

Since the multilayer configuration consists of a series connection where the second layer processes the output of the previous layer, they do not share any training steps. Consequently, the second layer requires its own hyperparameter study to optimise its performance. This cascaded design is illustrated in Figure 3.7 in Section (3.5).

4.2.1.1 Hyperparameter study of the second layer

The hyperparameter learning rate, number of epochs and n_0 determined for the first layer remain the same for usage of it as the intermediate layer, since the second layer has no effect on the first. However, the interconnected nature of the second layer requires an additional hyperparameter study where the learning rate and number of epochs of the second layer are individually defined. The learning rate variation is only performed on a model configuration that includes the laminar flow model for the first layer and the Adler-Hovorka literature model for the second layer. Due to the relative simplicity of this configuration and its ability to represent the main features of the model without introducing excessive complexity, it is chosen as the base case. While the investigation of the learning rate is limited to the base case, the optimal number of epochs is determined separately for each model configuration to locate its unique local minima and demonstrated in Appendix D.

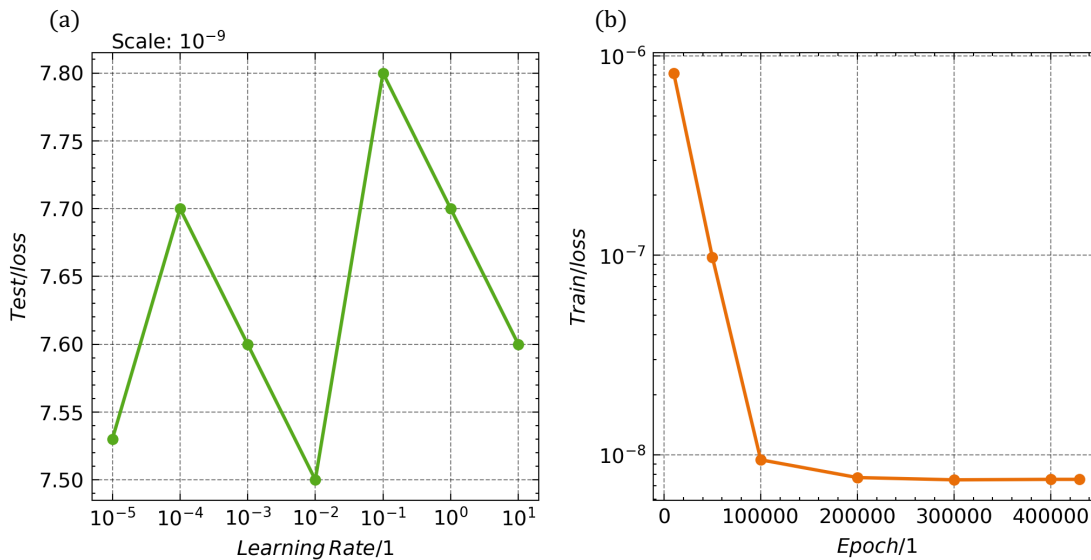


Figure 4. 8: The hyperparameter study of the configuration of the laminar flow and the Adler-Hovorka model for the second layer.

The Figure 4. 8.a illustrates that the second layer converges to its local minimum at a learning rate of 10^{-2} . While both layers reach their lowest loss values at the same learning rate, the second layer shows a more pronounced sensitivity to learning rate variations. In the range of 10^{-5} to 10^1 , the loss function exhibits wide variety, so the learning rate should be carefully tuned to avoid convergence to plateaus with higher loss values. Furthermore, according to the Figure 4.8.b the profile prediction of the second layer requires approximately 450,000 epochs to reach the local minima, which is significantly higher than the number of epoch used for the training go the first layer. The higher epoch requirement is attributed to the complex temporal dynamics of the input, which, unlike a simple step function with an instantaneous response, evolves up to 30 s to reach steady state in the laminar flow model. Thus, the gradual development of the

concentration profile introduces greater variability and complexity into the input data. Hence, the increased complexity in the temporal dynamics, in the form of a longer time for the concentration profile of the input to develop, requires the model to be trained for a longer period of time. In addition, in the implementation of the second layer, it is observed that both the kernel size and the output of the second layer increase compared to the first layer. The phenomenon is attributed to the mathematical dependencies of the second layer on the first layer. Since the output from the CNN is determined by Equation (3.2), which incorporates both the kernel and the input, this connected series leads to an increase in size. This increase results from the cascade configuration of the layers, where the output from one layer serves as the input for the next; thus, both output size and kernel size expand. Consequently, larger datasets require more epochs to adequately capture the underlying patterns [31-33].

The nRTD model, incorporating the Adler-Hovorka model in the first layer and the Cholette and Cloutier model in the second layer, is initially evaluated using the existing multilayer nRTD network architecture. Significant challenges are encountered during training, particularly in achieving convergence of the loss function to local minima. Initially, the multilayer nRTD configuration, that is used in the previous sections, is maintained for this model combination study. As shown in Figure 4.9, the resulting RTD function profile exhibited oscillatory behaviour, particularly around zero, close to steady-state behaviour. At the same time, the training profile, Figure 4.10, case 1, of this configuration showed that the plateau representing the local minima remained unreachable even after 800,000 epochs.

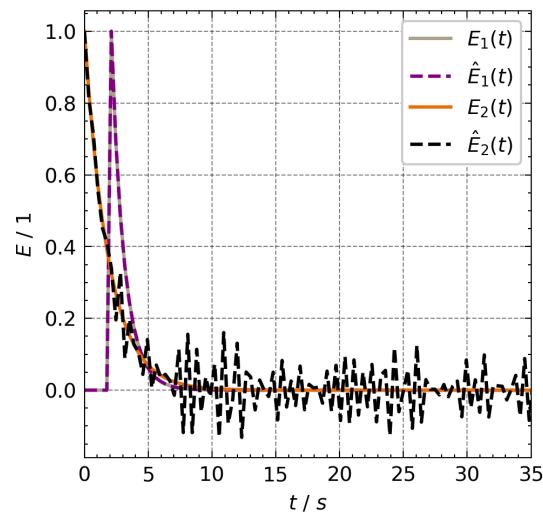


Figure 4. 9: The profile of Multi-Layered nRTD Modeling for Adler-Hovorka and Cholette-Cloutier Models with unrestricted E function.

It is argued that the observed oscillatory behaviour in the RTD function is caused by the model's attempts to adjust its predictions to compensate for negative values, which are physically unrealistic, thus preventing the model from reaching its convergence. Therefore, a modification

is introduced that restricts the weights in the model to a minimum value of zero to address this problem and match the physical constraint: $E(t)$ cannot be less than zero [5]. By implementing the non-negativity constraint, the focus of the model shifted from attempting to fit these oscillations to more efficiently capturing the essential dynamics of the system. As shown in Figure 4.10, case 2, the modified multilayer nRTD model with the constrained E-function achieved convergence to its local minima in approximately 600,000 epochs, demonstrating the effectiveness of the constraint in improving model performance and alignment with physical principles. The achieved RTD function is discussed in Section (4.2.1). However, it is important to note that while this modification proved beneficial for the complex combined model, it is not applicable for each model individually. Simpler models reach their minima more efficiently, and the constraint could limit the ability of the model to improve with additional epochs.

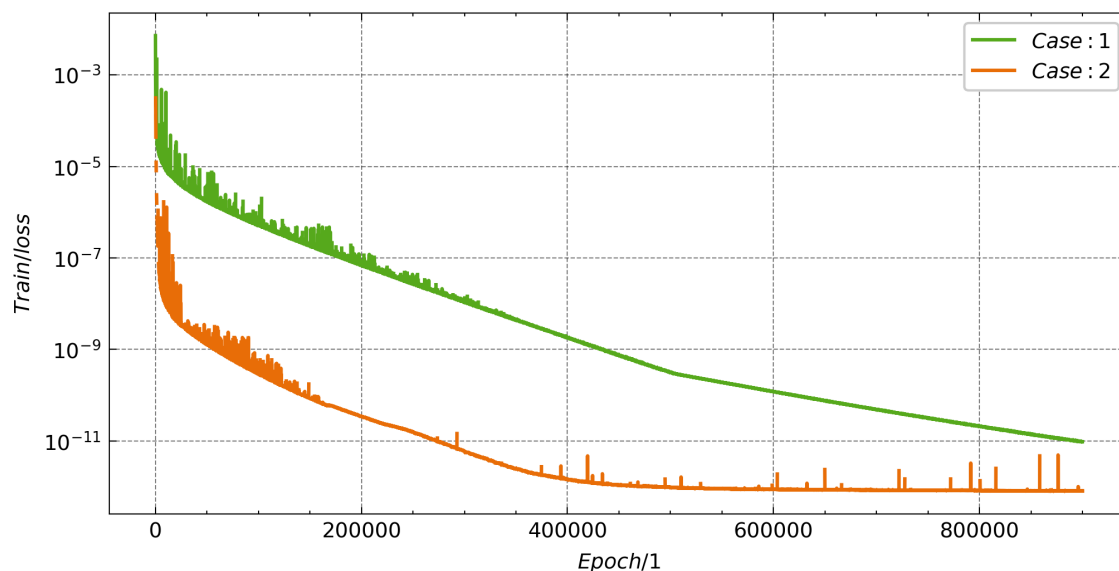


Figure 4. 10: Training profiles of the second-layer models: Adler-Hovorka and Cholette-Cloutier, demonstrating constrained and unconstrained $E(t)$ behaviours.

In order to validate the performance of the multi-layer nRTD model, an actual profile must be generated for error calculation during backpropagation, similar to the requirement for the first layer. However, data generation for the second layer is more complex than for the first layer, which has no preceding series connections. Consequently, two different approaches were investigated, as shown in Figure 4.11.a and Figure 4.11.b. These approaches utilise different rooted RTD profiles while maintaining consistency in the RTD function, using the theoretical E-function in both cases. The different dependencies allow to differentiate the performance of the branches derived from the literature-based model and the CNN-based model for loss calculation. The first approach, Figure 4.11.a, uses a concentration profile derived from pure theoretical data without any additional neural network effects. The second approach, Figure 4.11.b, involves the convolution of the theoretical RTD function with the concentration profile output from the first layer, which also serves as the input to the second layer.

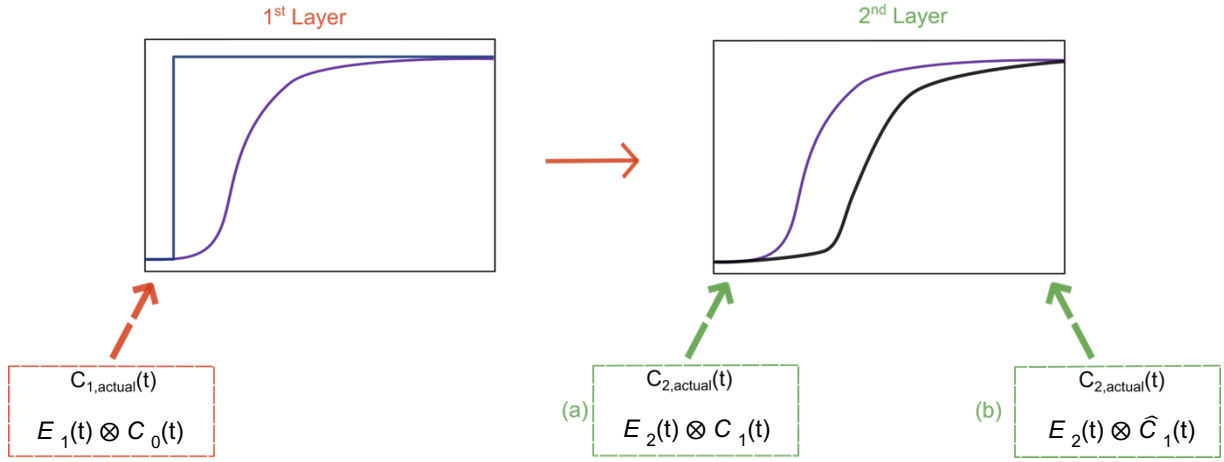


Figure 4. 11: Schematic representation of possible routes for actual profile dataset generation.

Mathematically, the first approach, represented by Equation (4.5), is based on a pure split path, distinguishing between the literature and CNN model concentration profiles. In this method, $C_{1,j}(t)$ is used for comparison and conventional sequential convolution is applied.

$$\hat{C}_{2,j}(t) = (\hat{E}_{2,j}(t) \otimes \hat{C}_{1,j}(t)) - (E_{2,j}(t) \otimes C_{1,j}(t)) \quad (4.5)$$

The alternative approach, which uses the $\hat{C}_{1,j}(t)$ for the convolution, combines the CNN result with $E_{2,j}(t)$ as shown in equation 4.6.

$$\hat{C}_{2,j}(t) = (\hat{E}_{2,j}(t) \otimes \hat{C}_{1,j}(t)) - (E_{2,j}(t) \otimes \hat{C}_{1,j}(t)) \quad (4.6)$$

The analysis indicates that better accuracy is achieved when the branches are purely split, although the resulting outputs are very close to each other. Since the data set resulting from equation 2 does not include any potential noise, the improved performance is observed, allowing the model to capture the details of the profile more effectively. The argument is consistent with the discussion presented in Section (4.1.3) where the advantages of noiseless data for feature extraction are highlighted.

4.2.1.2 Results and discussion of the second layer

Figure 4.12 indicates the capability of the nRTD for the multiple layered 1D convolution with varied compartment model configurations as first and second layer. The first row of the Figure 4.12 represents the combination of a basic and a complex compartment model, as the laminar flow model is implemented as an intermediate layer, while the Adler-Hovorka model is introduced to complement the overall system in terms of nonideal flow dynamics. Hence, the

nRTD modelling capability is extended to capture a wider range of connected flow behaviors by considering additional mixing behaviours with nonideality. Figures 4.12 (c-d) investigates the interaction between two complex compartment models, using the Adler-Hovorka model in the first layer and introducing its predicted profile as input to the second layer, which uses the Cholette and Cloutier model. In each case, the predicted concentration profiles and RTD functions show strong agreement with the simulated data throughout the time domain. The correspondence indicates the capability of the model to capture different flow behaviours and mixing patterns in multi-compartment systems, and demonstrates its application to complex multi-compartment RTD analyses. Furthermore, the strong agreement between predicted and simulated data regarding concentration and RTD profiles for the second layer, despite using a non-ideal step function, demonstrates the adaptability of the model to varying input functions. The performance indicates that the model is not restrictive to specific step input functions but is capable of effectively processing diverse, dynamic inputs. The successful utilisation of $\hat{C}_{1,j}(t)$ as an input to the second layer, which deviates from an ideal step function, confirms the model's flexibility in addressing time varying concentration profiles in the convolution. Figure 4.12 (a-b) shows the ability of the nRTD model to capture the characteristics of the Adler-Hovorka model with the corresponding laminar flow model's output from nRTD. In particular, a PFR with the back mixing through the STR compartment in the second layer, when $\tau_m = 1$ s, is represented by the RTD functions. Moreover, the both $E_{2,Adler}(t)$ and $\hat{E}_{2,Adler}(t)$ show a prolonged tail after a sharp initial rise, indicating the gradual release of tracer molecules with extended residence times, as the model theoretically includes. Therefore, nRTD effectively represents a complex system consisting of a laminar tube, PFR, perfectly mixed STR and stagnant zone without the necessity of separate formulations for each component. An illustration of the merging of two complex compartment models connected in series is exhibited in Figure 4.12 (c-d). Accordingly, the RTD functions which are $E_{2,Cholette}(t)$ and $\hat{E}_{2,Cholette}(t)$ retains the bypass behaviour of the system, consistent with the Cholette and Cloutier characteristics, despite being fed by the nRTD output, $\hat{C}_{1,Adler}(t)$, of the complex compartment model. Consequently, $\hat{E}_{2,Cholette}(t)$ represents a system of elements, a STR with a dead zone and a bypass flow which passed a PFR and a STR with a stagnant zone.

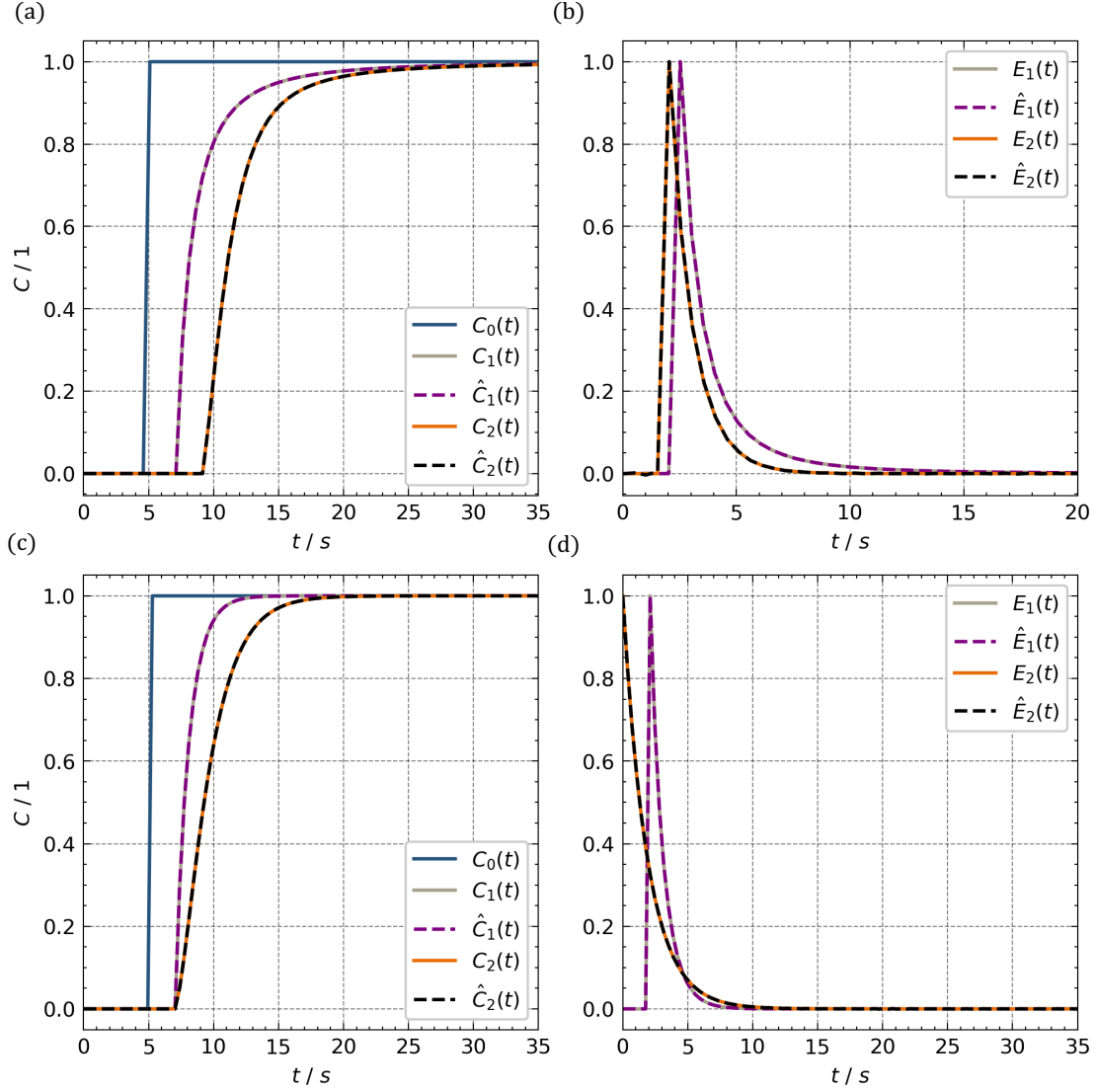


Figure 4. 12: The profile of Multi-Layered nRTD Modeling for (a-b) Integrated Laminar Flow and Adler-Hovorka, (c-d) Adler-Hovorka and Cholette-Cloutier Models.

4.3 Multilayer nRTD experimental implementation

The nRTD model demonstrates a strong reliability and accuracy in RTD analysis, as indicated by the comprehensive hyperparameter study and systematic implementation of literature compartment models. The successful integration of individual models in the first layer and their subsequent combination in the second layer confirms the diversity and effectiveness of the model. This thorough proof of concept process verifies that the model is suitable for application to real experimental data. Subsequently, the nRTD model is evaluated with experimental data from previous work to determine its applicability to practical applications and assess its predictive capabilities against empirical data [35].

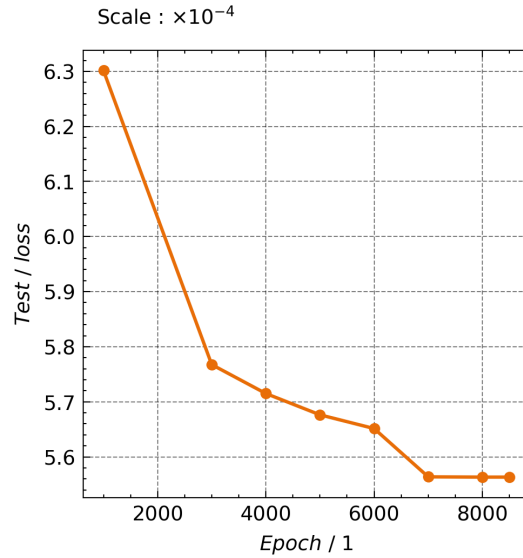


Figure 4. 13: The responses of the model according to different epoch numbers.

The hyperparameter investigation of the experimental data initiates the analysis. A part of the hyperparameter determination for the analysis of the experimental data is based on the previous study, which did not implement the two-layer system. Therefore, the learning rate of 10^{-3} based on previous results is preserved for the first layer [35]. However, the number of epochs requires reinvestigation due to the modified use of the data set in position 1. This adjustment is due to the interconnected character of the two-layer system, which imposes new temporal constraints on data processing, as explained in the methodology section. The reconfiguration of the temporal framework affects the training process of the model, potentially changing the optimal number of training iterations necessary for optimal convergence due to a shorter data length. Figure 4.13 illustrates the relationship between the number of epochs and the test/loss. The lowest tracer concentration is selected as the base case. It is used as the basis for the epoch number variation study for the first layer, which varies between 1000 and 8500 epochs. The results indicate a sharp decrease in test/loss up to 7000 epochs, followed by a relatively gradual decline in the epochs between 7000 and 8000. The profile appears to converge to a local minimum of around 8000 epochs; thereafter, only minimal improvement is observed. Hence, the optimal number of epochs for the first layer of the nRTD model for the experimental data is 8000.

Figure 4.14 shows the hyperparameter optimisation for the second layer, using data from position 2. The learning rate evaluation shown in Figure 4.14.a shows that the model achieves optimal accuracy at relatively low learning rates. A moderately high-performance plateau is reached between 10^{-3} and 10^{-1} , with the lowest test/loss value occurring at a learning rate of 10^{-1} . Accordingly, the learning rate of 10^{-1} is selected as the optimal value for training the second layer of the nRTD model with experimental dataset. The epoch number analysis for

the second layer, shown in Figure 4. 14.b, indicates that a higher number of epochs is required to reach the local minimum convergence. Therefore, increased complexity is attributed to the use of the first layer as input to the second layer, resulting in a more complicated input function. Based on the observed performance, an epoch number of 30,000 is chosen as the optimal value for the second layer. Although a slight improvement is achieved with further increases, this value offers a compromise between the need for model convergence and computational efficiency, ensuring optimal performance within the new temporal structure of the two-layer system [37].

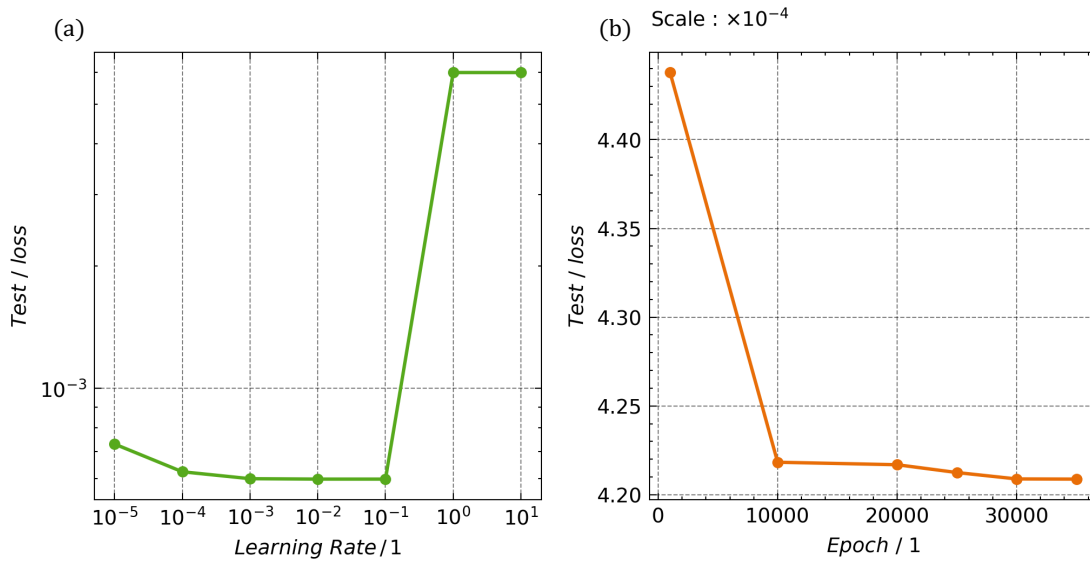


Figure 4.14: Hyperparameter optimisation for the second layer, using data from position 2.

Figure 4.15 demonstrates the applicability of the nRTD model to the experimental data, illustrating three different concentration values. The top row displays the highest tracer concentration at the lowest volumetric flowrate, while the lowest concentration is shown at the bottom with the highest flowrate in the system. The first column shows the concentration profiles at the first and second positions. Also, the RTD functions predicted by the model are plotted alongside the concentration profiles, respectively. The same approach that implemented in the Section (4.2.1) is used due to the complexity of the second layer and obey the theory of the RTD function, thus, the $\widehat{E}_1(t)$ is constrained to non-zero values [1].

The investigation of the model with varying concentrations at each point begins with the RTD analysis of the capillary, which serves as a representation of a single compartment. The step function is implemented as the input function for the first layer to match the experimental methodology at position 1. Subsequently, a strong agreement is achieved between the predicted concentration profile from the nRTD model, $\widehat{C}_1(t)$ and the experimental concentration profile, $C_1(t)$ at position 1 for both the highest and intermediate concentration studies.

Moreover, the concentration profiles, including both $\widehat{C}_1(t)$ and experimental data $C_1(t)$, converge to the steady state of the $C_0(t)$, demonstrating the ability of the model to accurately capture both the dynamic and steady-state behaviour of the system. The sudden increase in output concentrations at precisely $t = 1$ s indicate an almost instantaneous response of the compartment, reflecting the MS capillary's RTD and its lack of dead volume. The characteristic of the concentration profiles results from the pronounced decay of the $\widehat{E}_1(t)$. The early peak in the predicted RTD function indicates that certain portions of the fluid are taking shorter paths through the capillary, effectively bypassing the main flow. Additionally, the prolonged tail in the overall RTD functions is observed for both Figures 4.15 (b) and 4.15 (c). Several factors can contribute to uncertainty in the RTD profile of the capillary. In the experimental setup, a few centimetres of pipe between the switching valve and the capillary can allow pressure fluctuations to propagate, potentially affecting flow characteristics before reaching the capillary [34]. Additionally, the switching process typically causes a small pressure shock that varies with flow rate and influences the flow dynamics through the capillary. Possibly, the pressure at the inlet of capillary might impact the flow rates, thus the flow regime. As shown in Figure 4.15.e, the convolved results at higher flow rates indicate a decrease in tracer concentration and an underfitting of the concentration profile, $\widehat{C}_1(t)$. Furthermore, the convolved results at higher flow rates, where tracer concentration decreases, exhibit an underfitting of the concentration profile, as shown in Figure 4.15.e. Figure A.1, which shows the experimental data at each point with different flow rates in different sub-plots, indicates that this arises from a temporal discrepancy in data acquisition rather than a possible inherent flow dynamic. Data acquisition is initiated earlier than the actual introduction of the tracer, resulting in a shift in the observed concentration profiles. Thus, the step increase in tracer concentration appears in the experimental results later than expected instead of at $t = 1$ s as assumed in the nRTD model. Therefore, an extended period of baseline concentration is followed by a delayed increase. As a result of the mismatch, the model struggles to accurately fit the initial slope of the concentration curve, specifically at the beginning of the recorded data. Consequently, the E -function of the first position at the highest flow rate could not be observed as a complete profile; instead, only a tail is visible, as seen at other flow rates at position 1. This observation highlights the critical importance of accurate timing in experimental procedures and systems with rapid dynamics where minor temporal discrepancies can significantly affect data interpretation and model fitting. Moreover, an additional factor contributing to the described underfitting profile may stand due to the lacking temporal coverage of the kernel and input function in the model, as outlined in methodology Section (3.4). The limited time frame may not allow the full development of the tracer concentration profile, particularly in high-flow scenarios involving

rapid dynamics. Consequently, the ability of the model to accurately capture the whole concentration curve is compromised, leading to underfitting.

The second position in the analysis demonstrates the performance of the model for a more complex system response, encompassing not only the RTD analysis of one compartment but additionally the time the tracer spends in the MS capillary and the tube preceding the reactor. Within this second layer, $\widehat{C}_1(t)$, serves as the input, building upon the results obtained in the initial layer. Despite the increased complexity introduced by two compartments in the second layer, the model maintains its capacity to effectively capture the system dynamics, as evidenced by the $\widehat{C}_2(t)$ following the general trend of $C_2(t)$. However, the predicted profiles do not precisely match the experimental data. Instead, they present a smoothed appearance. This deliberate smoothing effect of the model shows avoiding overfitting to noise in the data. In accordance with the discussion in Section (4.1.3), this allows the model to learn and generalise the actual underlying signal from noisy data rather than fitting to random fluctuations. However, despite this averaged behaviour in the predicted concentration profiles, Figure 4.15.c reveals an underfitting phenomenon between $t = 1$ s and $t = 2$ s for $\widehat{C}_2(t)$. This underfitting is comparable to the case observed at the highest flow rate in the first layer, where data recording commenced before the intended time point. Consequently, the nRTD model predicts a concentration profile with an immediate response, overlapping with the first layer's concentration profile. The output of the nRTD model is based on profiles that initiate before $t = 2$ s, leading to a misalignment between the predicted and actual RTD characteristics of the compartment. Subsequently, the $\widehat{E}_2(t)$ of the intermediate flow rate differs from the functions observed at the highest and lowest flow rates examined in the study where the RTD profiles for the highest and lowest flow rates exhibit a delayed peak followed by a tail. This suggests that the compartment can displays laminar behaviour internally, which this characteristic could partially define. Also, comparing Figure 4.15.f with Figure 4.15.b, it is evident that the RTD profile is slightly narrower. This observation is consistent with literature that indicates increasing flow rates generally lead to narrower RTD tails. Alternatively, the narrow $E(t)$ profiles can be interpreted as suggesting that the second compartment experiences some degree of axial dispersion. Therefore, the RTD profiles obtained from the experimental set-up are complicated to compare and define with the existing compartmental models because of their uniqueness and non-ideality. Despite these complexities, the nRTD model effectively characterises the system by accurately capturing concentration profiles and performing RTD analysis without requiring preliminary mathematical modelling or subsequent iterations.

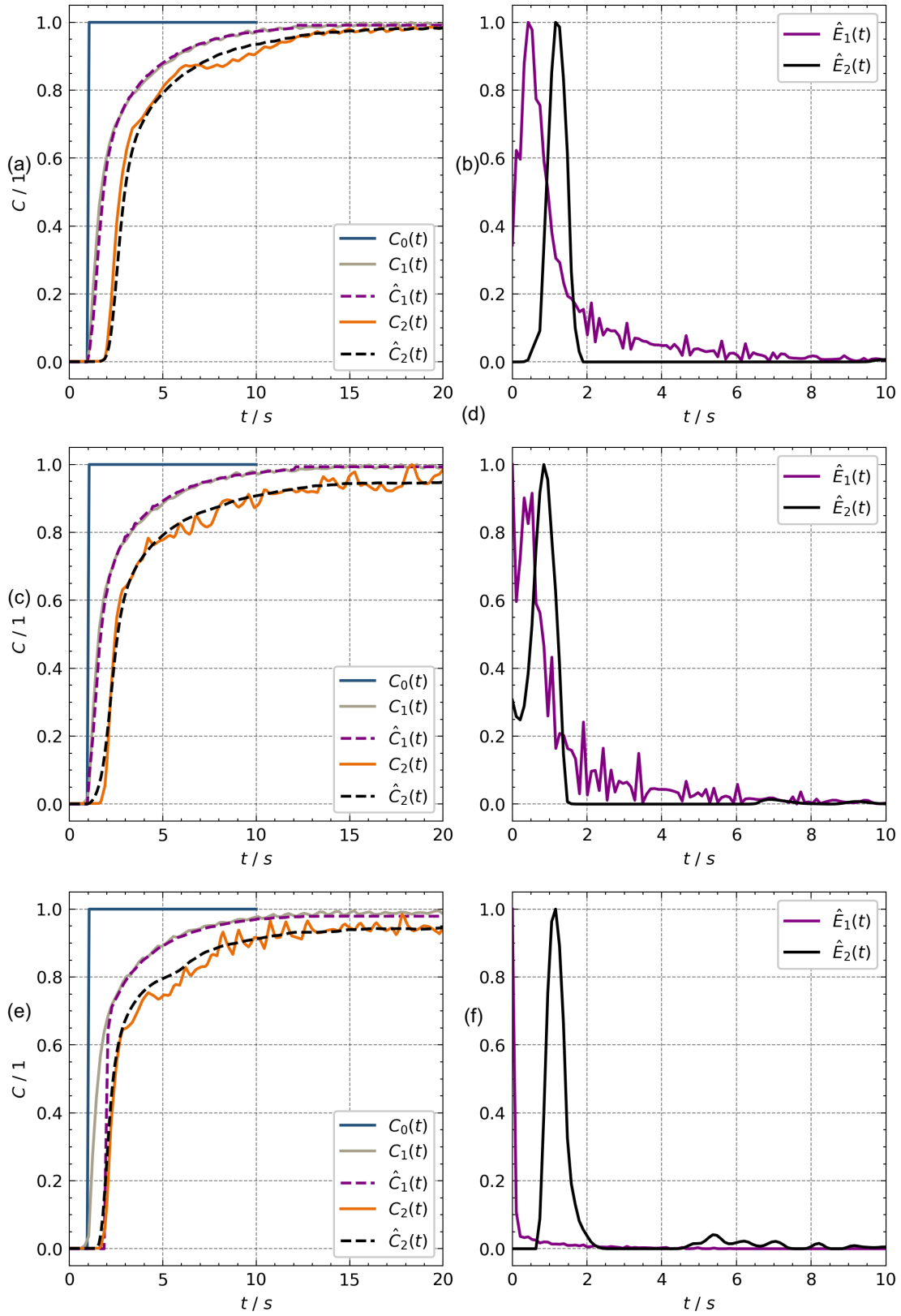


Figure 4.15: The profiles of Multi-Layered nRTD Modeling for experimental data at flowrate (a-b) 100 mL m⁻³, (c-d) 150 mL m⁻³ and, (e-f) 200 mL m⁻³

5 Conclusion

The RTD is an important phenomenon that describes the distribution of times that fluid particles spend within a system. Its analysis is essential in chemical reaction engineering, particularly for understanding fluid dynamics in dynamic processes. Despite its significance, conventional RTD analysis methods, compartment models, can be complex and time-consuming due to their iterative nature with oversimplification regarding the flow dynamics. This study developed a novel approach to RTD analysis by incorporating CNN to address the limitations of conventional methods. The aim is to enhance the evaluation of non-idealities in flow dynamics, especially in complex systems. By targeting the compartmental method, this new analysis technique seeks to eliminate unrealistic assumptions of conventional RTD analysis and provide a more accurate representation of fluid behaviour.

The present study focuses on the development of a flexible and robust neural network residence time distribution (nRTD) model, which is optimised through hyperparameter investigations on learning rate, epoch number and discretization of the output data. Theoretical datasets are generated from literature based compartment models to create synthetic data for training. The theoretical models examined range from basic compartment models to complex compartment models that account for non-idealities, including bypassing, stagnant regions, and dead zones. The basic compartment models analysed include laminar flow and dispersion models, along with more advanced compartmental frameworks which are Cholette and Clotier, Adler and Hovorka, as well as the Unified time delay model. The selected models are deliberately chosen to encompass a diverse range of compartmental mechanics and system behaviours, facilitating a comprehensive analysis that links computational results with experimental configuration. Additionally, a realistic test rig setup utilizing the PTK method is used for the validation of the capability of the nRTD model.

In the presented work, network construction is based on mathematical equality between RTD analysis and, CNN, where both fundamentally depend on the convolution process. Therefore, with the absence of a bias term, the RTD function can be represented with the weights of the 1D convolution layer term in the CNN. In this approach, the convolution layer convolves the input data with the currently predicted kernel as it slides to obtain the predicted output from the model. The forward and backward passes in the network are iterated until the specified number of epochs in the model. Through the iterative refinement, which mimics a deconvolution operation without explicitly implementing it as a separate layer, the model implicitly learns the inverse operation, effectively performing deconvolution by adjusting its parameters to minimise the discrepancy between predicted and actual outputs. Thus, it is able to invert the convolution process through parameter optimisation rather than explicit algorithmic implementation.

The nRTD model, consisting of a 1D convolutional layer, is validated via the theoretical compartment models to provide a proof of concept for further investigation. According to the hyperparameter studies, the investigations are carried out with a learning rate of 10^{-2} for the layer and varying the number of epochs from 17,000 to 35,000 as a function of the model. The increase in the epoch number is observed for both broader RTD profiles and inherently complex RTD functions. The nRTD model demonstrates its ability to capture a wide range of flow dynamics, from purely convective behaviour represented by laminar flow models to the transition between strong and weak backmixing scenarios in dispersion models. Moreover, the non-ideal flow behaviour is explicitly captured and specifically for the model proposed by the Cholette and Cloutier model which takes into account bypassing and dead volumes, the pronounced decay in the inlet proves the captured behaviour. Therefore, unlike conventional RTD methods that rely on predefined models and extensive manual tuning, the nRTD framework learns directly from the data and adapts to different flow scenarios and non-idealities.

In order to extend the utilisation range of the nRTD model, the introduction of multiple convolution layers is investigated. The multi-layer nRTD model is implemented as a cascade workflow with two one-dimensional convolution layers connected in series, where the predicted output of the first layer serves as the input for the second layer. This configuration is examined in two scenarios. Firstly, the combination of basic models that are the laminar flow model and Adler-Hovorka, and then a combination of two complex models as Adler-Hovorka and Cholette and Cloutier, first and second layer respectively, is studied. The increased complexity of the input, $\hat{C}_{1,j}(t)$, which differs from an ideal step function to a broader profile that develops over time, requires a longer training period for the model to accurately capture the RTD. Consequently, the second layer requires a relatively longer training period. This observation overcomes the initial presumption of model dependency on ideal inputs and instead demonstrates that the nRTD model can effectively utilise different input profiles while maintaining accurate predictions of the target concentration profile. This accuracy is achieved by precise prediction of the $E(t)$ during the convolution process.

The multilayer nRTD model is validated using the experimental work performed in the previous study [34]. Hyperparameter optimisation is performed for both layers due to the modified use of experimental data. To obtain predicted RTD functions for both layers, a compromise is made by shortening the time domain, resulting in insufficient t_e capture the full RTD profiles of the individual compartments. Therefore, deviations in the RTD profiles of the second layer are observed, mainly due to its dependence on the predictions of the first layer and limitations in both the model and the experimental data. The predictions of the model for RTD functions are

analysed to elucidate potential flow dynamics, taking into account non-ideal flow characteristics. Analysis of the experimental data reveals possible flow behaviours in the two compartments of the system. The first compartment, the MS capillary, exhibits a bypass behaviour analogous to that observed in the Cholette and Cloutier model in theoretical compartmental studies, with an instantaneous response of RTD profile. In the second compartment, a possible degree of axial dispersion in the system is defined. However, the RTD profile of this compartment is also similar to laminar flow models, making a definitive classification based on existing compartment models in the literature difficult according to the RTD profile is difficult. This ambiguity underlines the complexity of real-world fluid dynamics and the limitations of conventional modelling approaches. Therefore, the data-driven nRTD model is a potentially effective tool for the analysis of complex multi-compartment systems, where conventional approaches often struggle to describe the system with adequate accuracy.

A Appendix

Appendix A contains the detailed representation of the training data sets used in section (4.3). The experimental data are obtained in the previous study via the experimental setup demonstrated for PTK experiments [34,35].

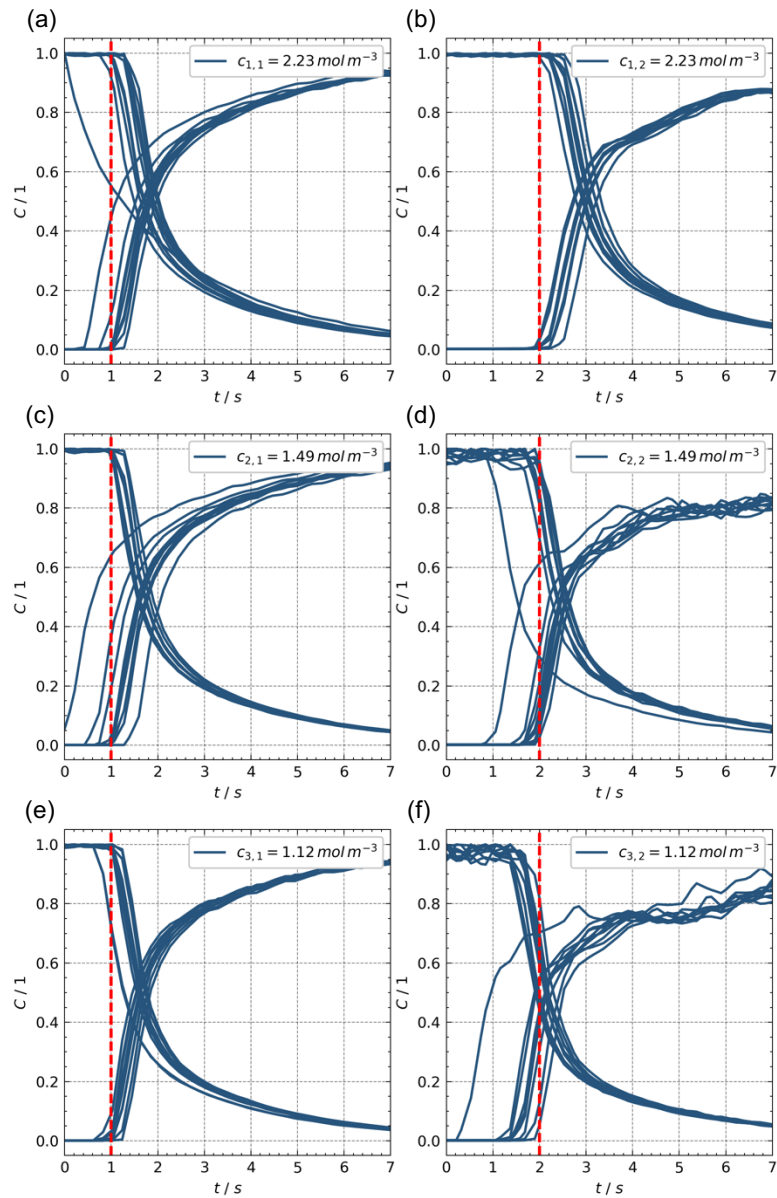


Figure A. 1: Development of the normalised concentration profiles for experimental data at flowrate (a-b) 100 mL m^{-3} , (c-d) 150 mL m^{-3} and, (e-f) 200 mL m^{-3}

B Appendix

The theoretical data generation associated with the implementation of the Adler-Hovorka model, using the inverse Laplace transform with SymPy in Python is presented. The detailed explanation regarding the implementation can be found in Section (3.3)

```
def compute_inverse_laplace(coefficients, t_values):
    # Define symbols for Laplace transformation (s) and time domain (t)
    s, t = sp.symbols('s t')
    # Extract alpha value from the coefficients dictionary
    alpha = coefficients['alpha_val']
    # A list to store results for each parameter combination
    results = []
    # Loop over each value of tau_a
    for tau_a_val in coefficients['tau_a_val']:
        # Loop over each value of tau_p
        for tau_p_val in coefficients['tau_p_val']:
            # Loop over each value of beta
            for beta_val in coefficients['beta_val']:
                # Compute tau_m based on alpha and beta
                tau_m_val = (beta_val * (1 - alpha)) / alpha
                # Define the Laplace transform function F(s) using given parameters
                F_s = (sp.exp(-tau_p_val * s)) / (1 + beta_val + tau_a_val * s - (beta_val / (1 +
tau_m_val * s)))
                # Compute the inverse Laplace transform of F(s) to get f(t)
                f_t = sp.inverse_laplace_transform(F_s, s, t)
                # Convert symbolic f(t) into a numerical function for evaluation
                f_t_numeric = sp.lambdify(t, f_t, modules="numpy")
                # Evaluate the function f(t) at the given t_values
                E_t = f_t_numeric(t_values)
                # Store the computed parameters and E(t) in the results list
                results.append({
                    'tau_a_val': tau_a_val,
                    'tau_p_val': tau_p_val,
                    'tau_m_val': tau_m_val,
                    'beta_val': beta_val,
                    'E_t': E_t,
                })
    # Compute the integral of E(t) over the t_values using numerical integration
```

```
        integral_E_t = np.trapz(E_t, t_values)
    # Return the list of results for all parameter combinations
    return results
# Example input coefficients dictionary
coefficients = {
    'tau_a_val': np.array([]),
    'tau_p_val': np.array([]),
    'beta_val': np.array([]),
    'alpha_val': }
```

C Appendix

Appendix C illustrates all related information mentioned and referenced in this thesis for the first layer implementation of the nRTD model. The associated model parameters and hyperparameters for each compartment model are given. In addition, the profiles obtained from parameter variation are also illustrated in this section.

Table C. 1: Discretisation and temporal parameters for the nRTD model applied to the study of the dispersion model, $Bo = 1$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	300 s	150 s	150 s

Table C. 2: Discretisation and temporal parameters for the nRTD model applied to the study of the dispersion model, $Bo = 10$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	140 s	70 s	70 s

Table C. 3: Discretisation and temporal parameters for the nRTD model applied to the study of the dispersion model, $Bo = 100$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	50 s	25 s	25 s

Table C. 4: Discretisation and temporal parameters for the nRTD model applied to the study of the Adler Hovorka model, $\tau_a = 1$ second.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	70 s	35 s	35 s

Table C. 5: Discretisation and temporal parameters for the nRTD model applied to the study of the Adler Hovorka model, $\tau_a = 3$ seconds.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	70 s	35 s	35 s

Table C. 6: Discretisation and temporal parameters for the nRTD model applied to the study of the Cholette and Cloutier model, $\beta = 0.1$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	50 s	25 s	25 s

Table C. 7: Discretisation and temporal parameters for the nRTD model applied to the study of the Cholette and Cloutier model, $\beta = 0.3$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	50 s	25 s	25 s

Table C. 8: Discretisation and temporal parameters for the nRTD model applied to the study of the Cholette and Cloutier model, $\beta = 0.9$.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	100 s	50 s	50 s

Table C. 9: Discretisation and temporal parameters for the nRTD model applied to the study of the Unified time delay model, $J = 2$, $\tau_s = 1$ second.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	134	67	60 s	40 s	20 s

Table C. 10: Discretisation and temporal parameters for the nRTD model applied to the study of the Unified time delay model, $J = 2$, $\tau_s = 3$ seconds.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	60 s	30 s	30 s

Table C. 11: Discretisation and temporal parameters for the nRTD model applied to the study of the Unified time delay model, $J = 2$, $\tau_s = 1$ second.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	134	67	60 s	40 s	20 s

Table C. 12: Discretisation and temporal parameters for the nRTD model applied to the study of the Unified time delay model, $J = 5$, $\tau_s = 3$ seconds.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$
Value	200	100	101	60 s	30 s	30 s

Model	Epoch number/ 1
Laminar flow model	17,000
Dispersion model, $Bo = 1$	18,000
Dispersion model, $Bo = 10$	25,000
Dispersion model, $Bo = 100$	30,000
Adler Hovorka, $\tau_a = 1$ second	33,000

Adler Hovorka, $\tau_a = 3$ second	33,000
Cholette and Cloutier, $\beta = 0.1$	23,000
Cholette and Cloutier, $\beta = 0.3$	22,000
Cholette and Cloutier, $\beta = 0.9$	16000
Unified time delay model, $J= 2$, $\tau_s=1$ second	28,000
Unified time delay model, $J= 2$, $\tau_s=3$ seconds	30,000
Unified time delay model, $J= 5$, $\tau_s=1$ second	21,000
Unified time delay model, $J= 5$, $\tau_s=3$ seconds	21,000

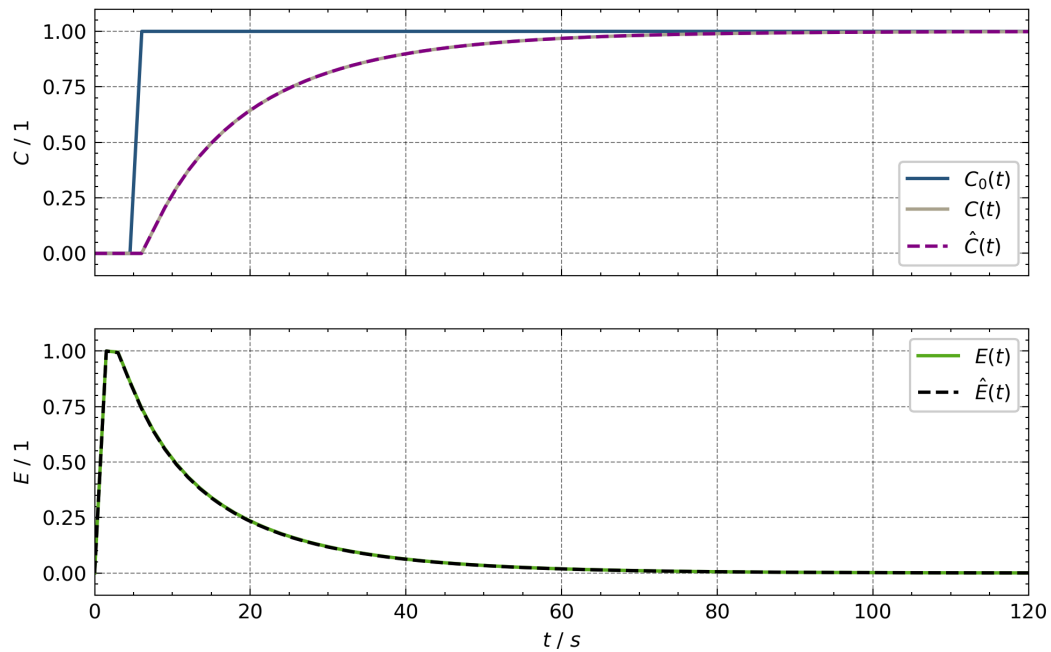


Figure C. 1: nRTD model output of the literature based dispersion model, $Bo=0.1$.

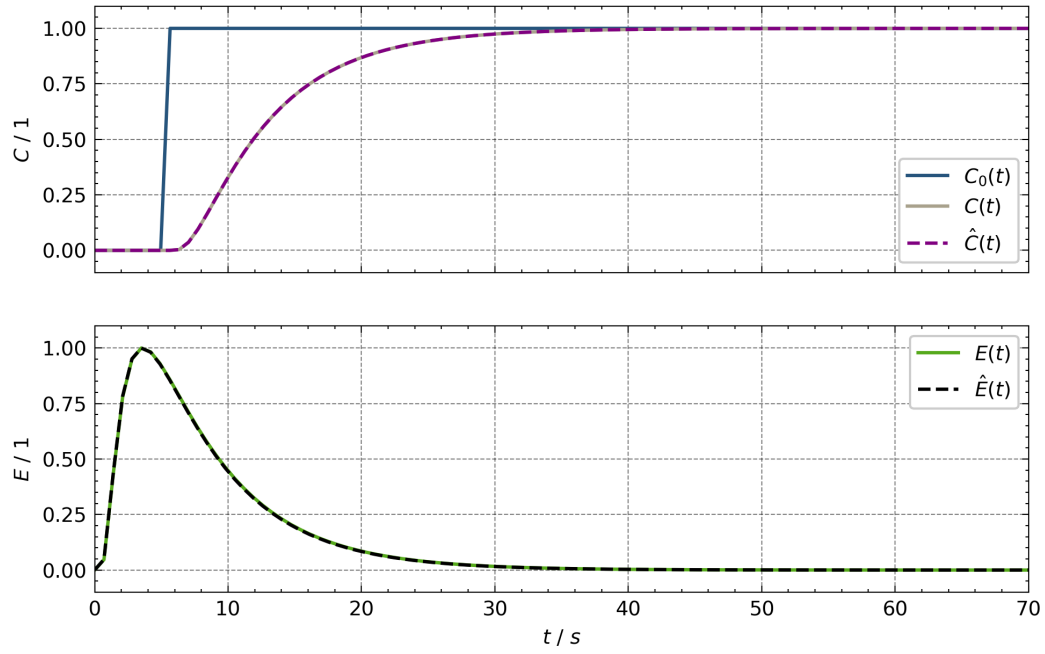


Figure C. 2: nRTD model output of the literature based dispersion model, $Bo=0.3$.

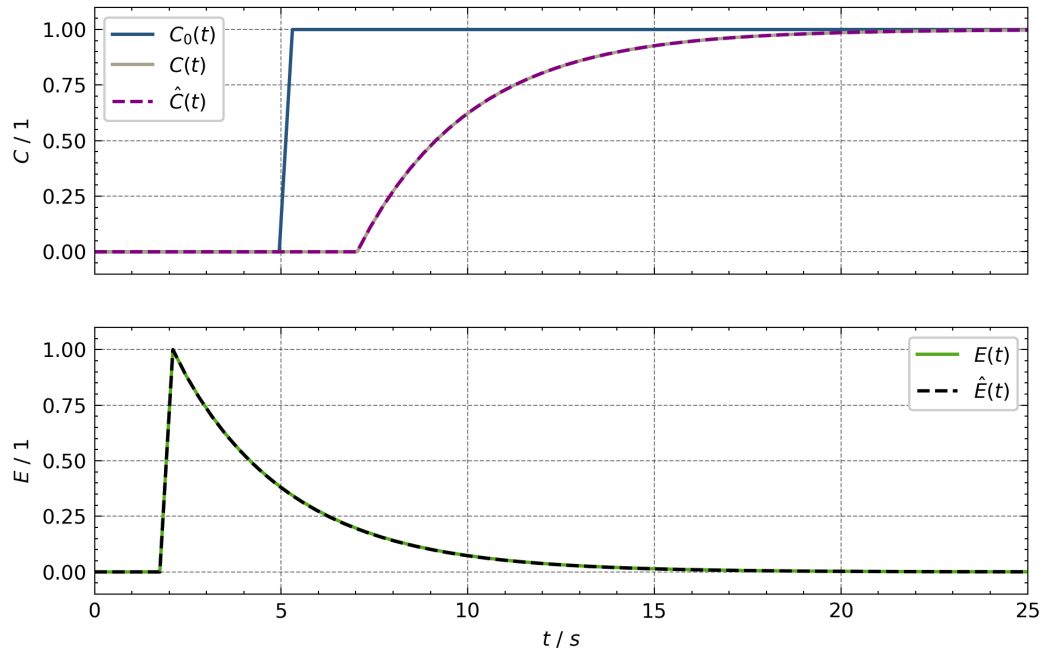


Figure C. 3: nRTD model output of the literature based Adler Hovorka model $\tau = 3$.

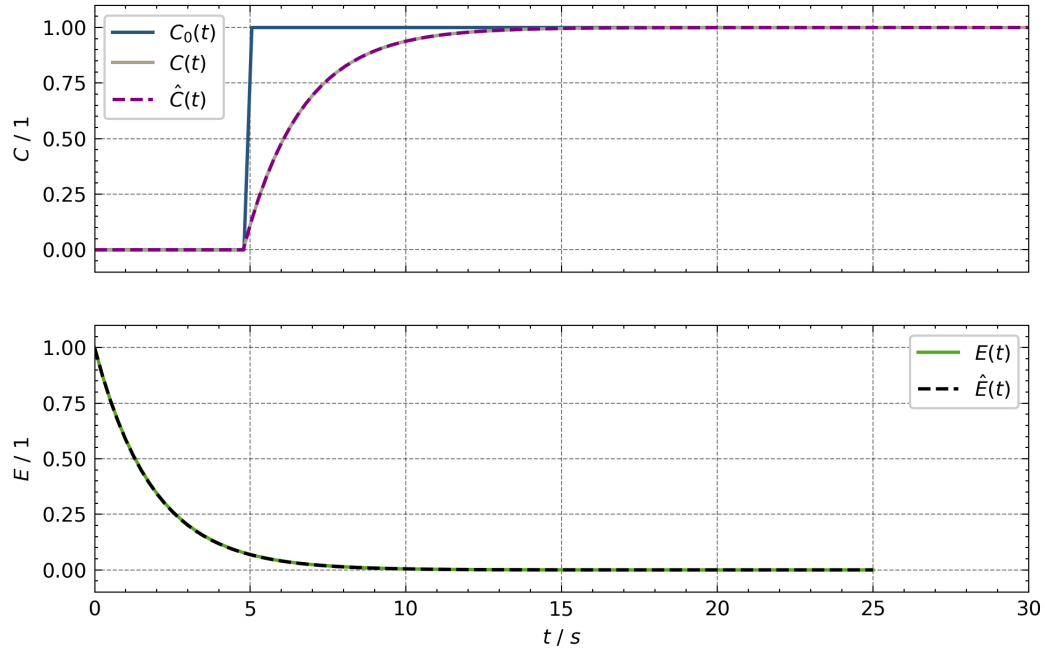


Figure C. 4: nRTD model output of the literature based Cholette Cloutier model $\beta = 0.3$.

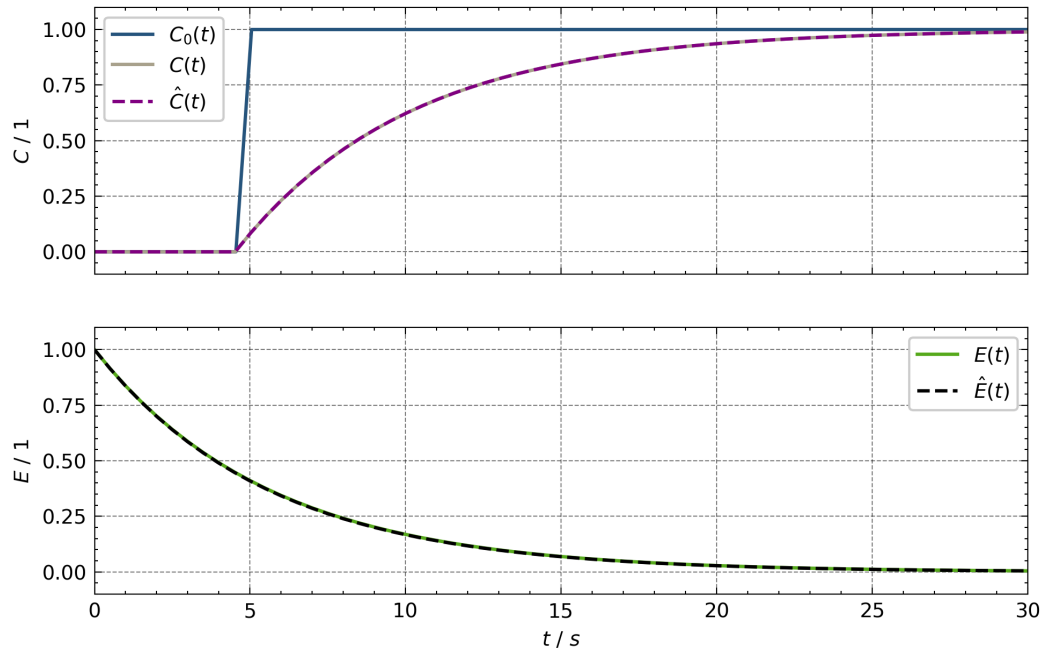


Figure C. 5: nRTD model output of the literature based Cholette Cloutier model $\beta = 0.9$.

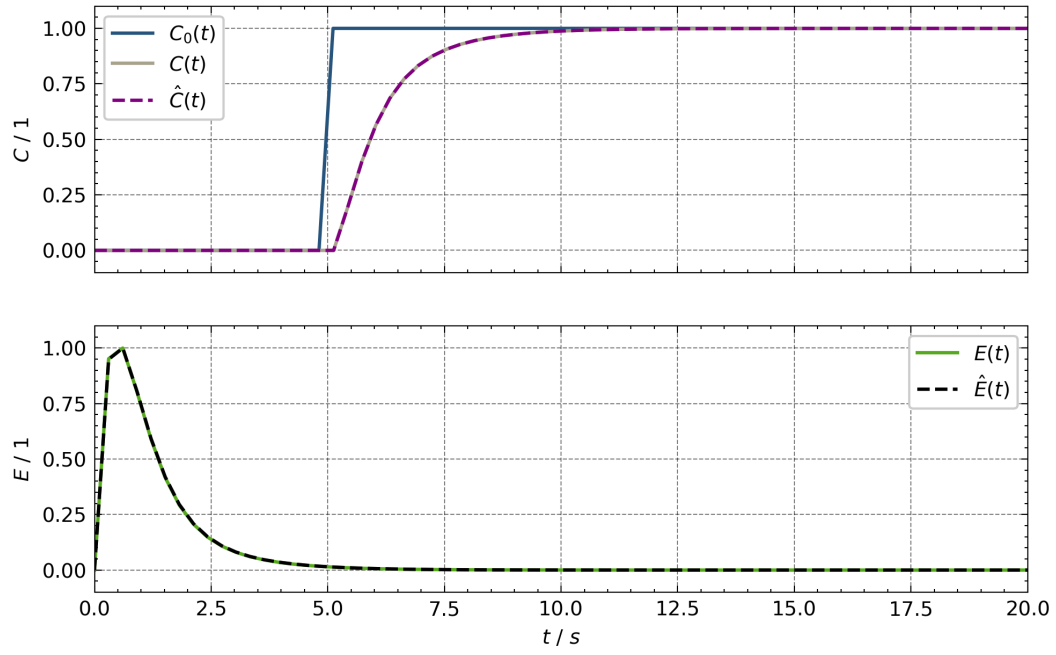


Figure C. 6: nRTD model output of the literature based Unified time delay model, $J= 2$, $\tau=1$ second

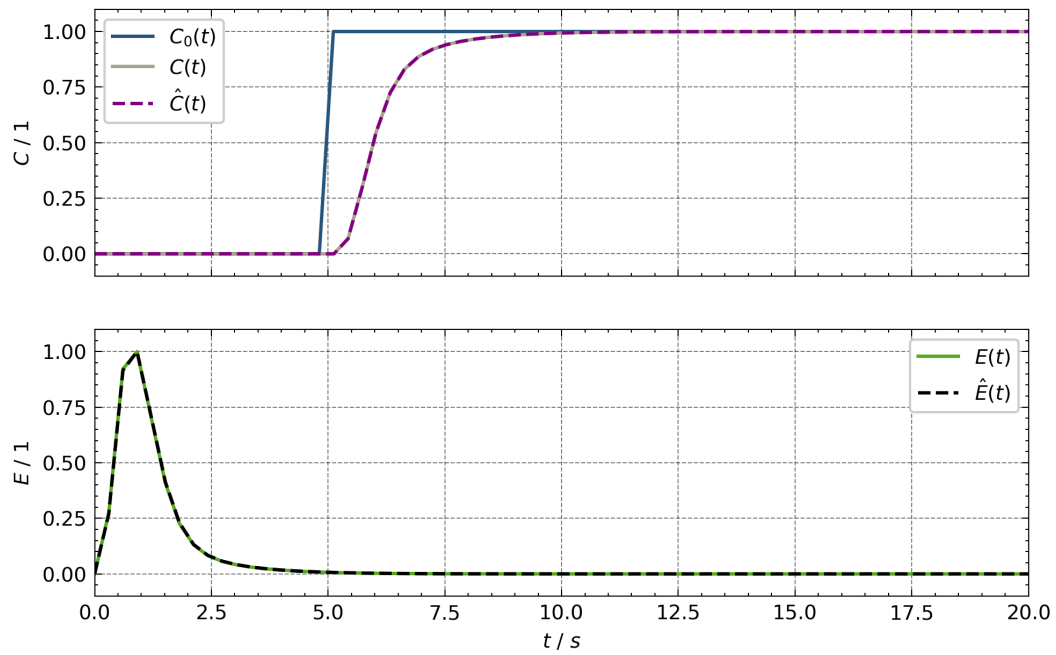


Figure C. 7: nRTD model output of the literature based Unified time delay model, $J= 5$, $\tau= 1$ second

D Appendix

The nRTD model parameter of the 2nd layer for Adler-Hovorka combined with Cholette Cloutier case study, is shown in Table D.1.

Table D. 1: Parameters of the 2nd layer convolution in the nRTD model: Adler-Hovorka with Cholette Cloutier model.

Parameter	$n_{o,1}$	$n_{i,1}$	$n_{e,1}$	$\Delta t_{o,1}$	$\Delta t_{i,1}$	$\Delta t_{e,1}$	$n_{o,2}$	$n_{i,2}$	$n_{e,2}$	$\Delta t_{o,2}$	$\Delta t_{i,2}$	$\Delta t_{e,2}$
Value	200	86	115	70 s	30 s	40 s	342	200	143	120 s	70 s	50 s

E Appendix

All relevant Python implementations explained in this thesis are available in the GitLab repository linked below. This includes the theoretical dataset generation for both the first and second layers, including the individual implementations of the nRTD model to compartment models. In addition, the repository contains implementations related to experimental data demonstrating real case applications.

<https://gitlab.uni-ulm.de/ciw-ag-g-ttel/private/nRTD.git>

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Statutory declaration

I hereby declare in lieu of an oath that I have completed this work independently. Thoughts taken directly or indirectly from external sources are marked as such.

The work has not yet been submitted to any other examination authority.

Ulm, 10th January, 2025

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Tuana Oyuncu

Declaration of Autonomy for written assignments, especially theses

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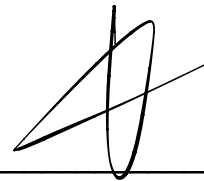
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